



Semiconductor IC Devices

Saroj Rout

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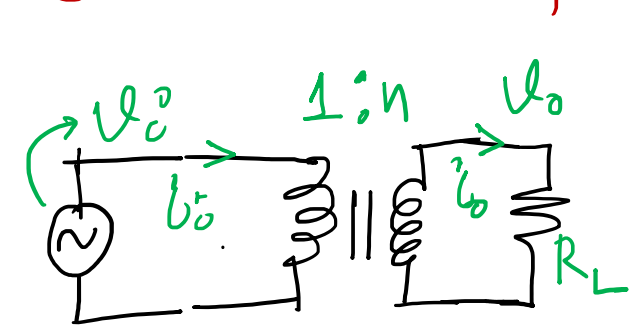
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Silicon
...beyond teaching

► Motivation for Semiconductor Device: Power Gain

① Can a transformer provide power gain?



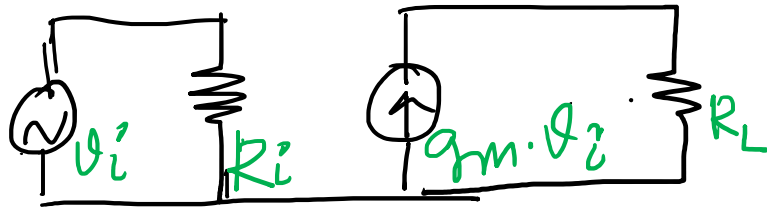
$$P_{out} = \frac{v_o^2}{R_L} = \frac{(n \cdot v_i)^2}{R_L}$$

$$\frac{v_i^2}{R_L/n^2} = P_{out}$$

$P_{in} = P_{out}$
NO POWER GAIN

$$P_{in} = v_i \times i_i = v_i \times n i_o = v_i \times n \cdot \frac{n \cdot v_i}{R_L} = \frac{v_i^2}{R_L/n^2} = P_{out}$$

② How about a transistor?



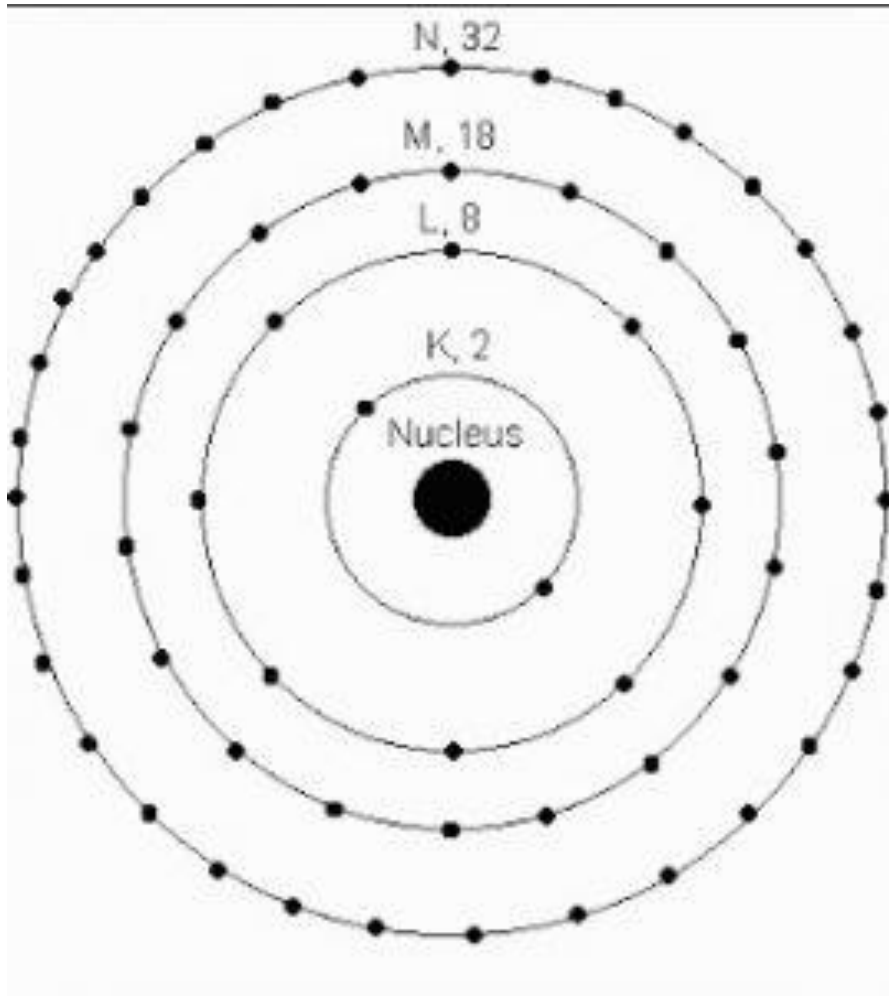
$$P_{out} = (g_m \cdot v_i)^2 \cdot R_L$$

$$P_{in} = \frac{v_i^2}{R_i}$$

$$\text{Power gain} = \frac{P_{out}}{P_{in}} = \frac{g_m^2 \cdot R_L \cdot R_i^2}{R_i}$$

Typically R_i is large & $g_m R_L$ is large as well

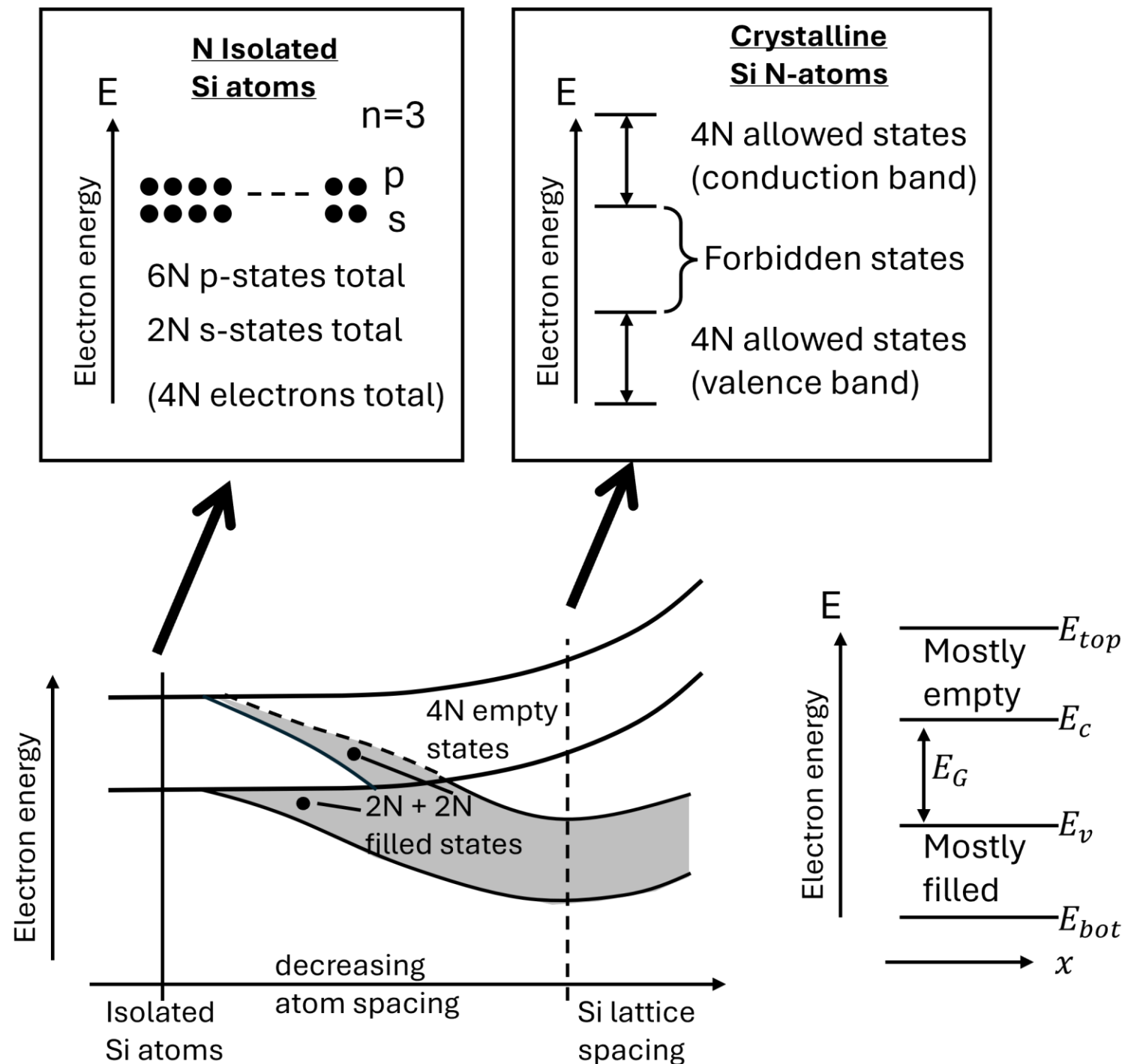
► Bohr's Atomic Model



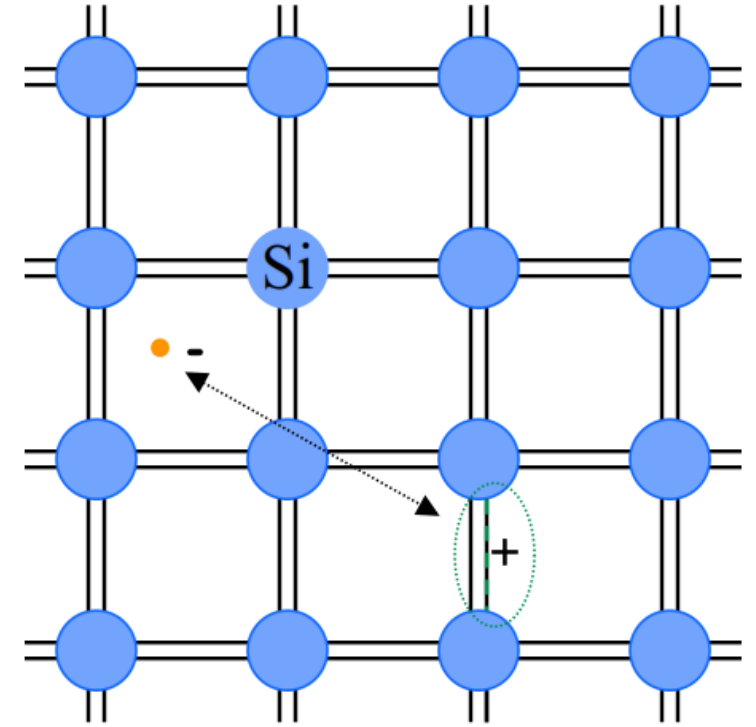
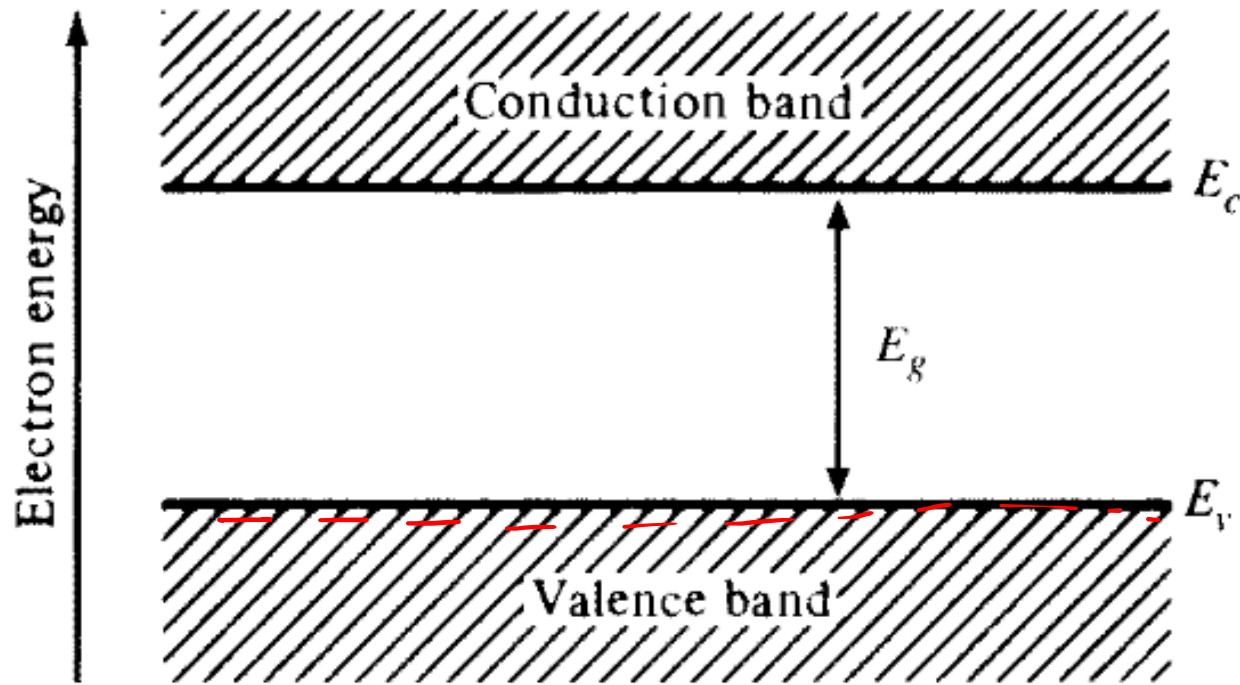
$$\text{Allowed States: } E_n = -\frac{E_B}{n^2}$$

$$\text{For Hydrogen Atom: } E_B = 13.6 \text{ eV}/n^2$$

► Energy Band Model



► Band Diagram / Intrinsic Semiconductor

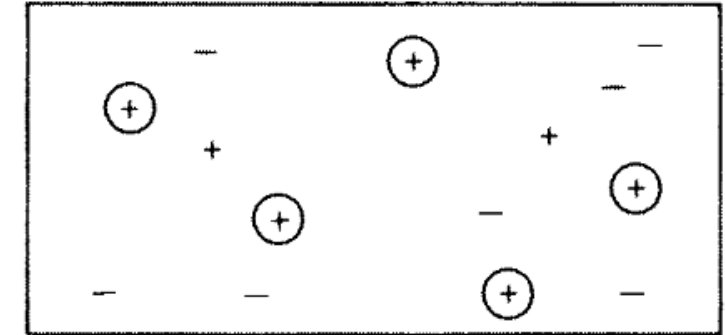
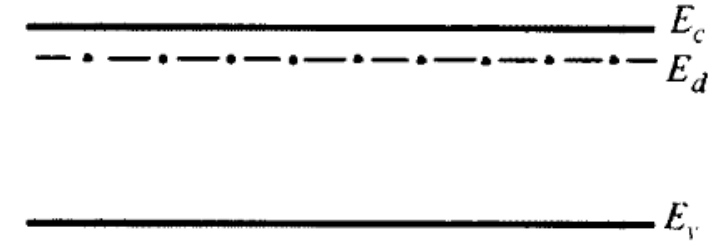
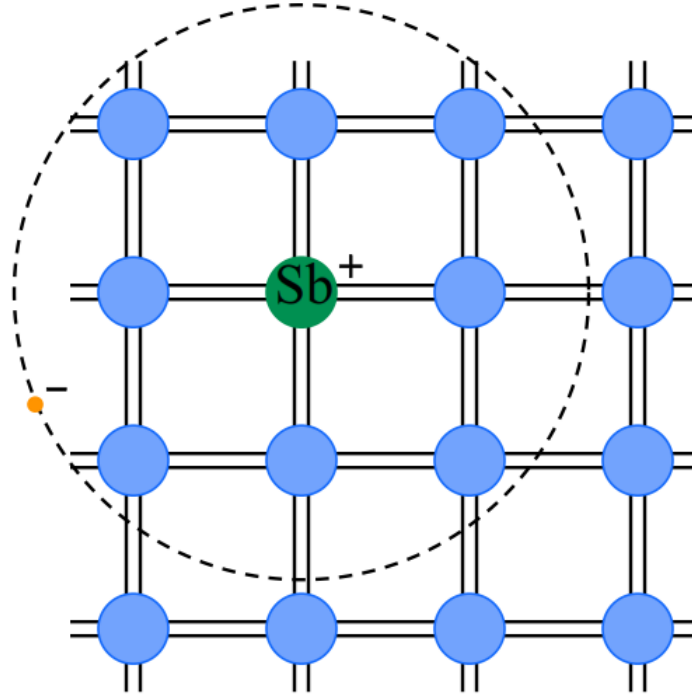


- Band diagram is a representation of the allowable energy states for electrons.
- Fermi Level : $P(\text{electron occupying empty state}) = 0.5$ (50%)

► Extrinsic Semiconductor n-type

sn)

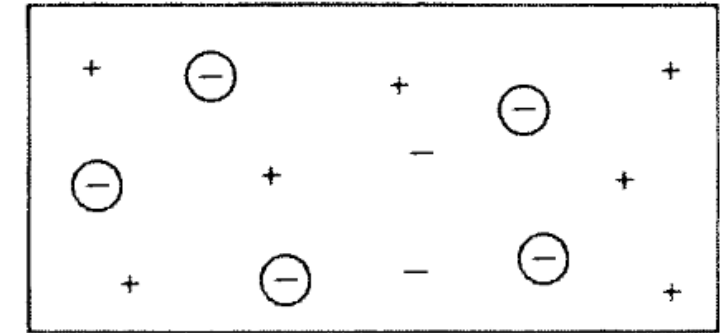
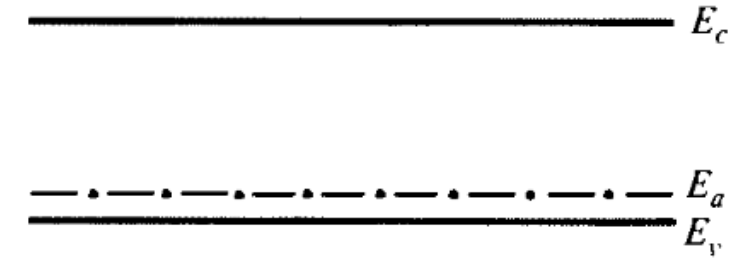
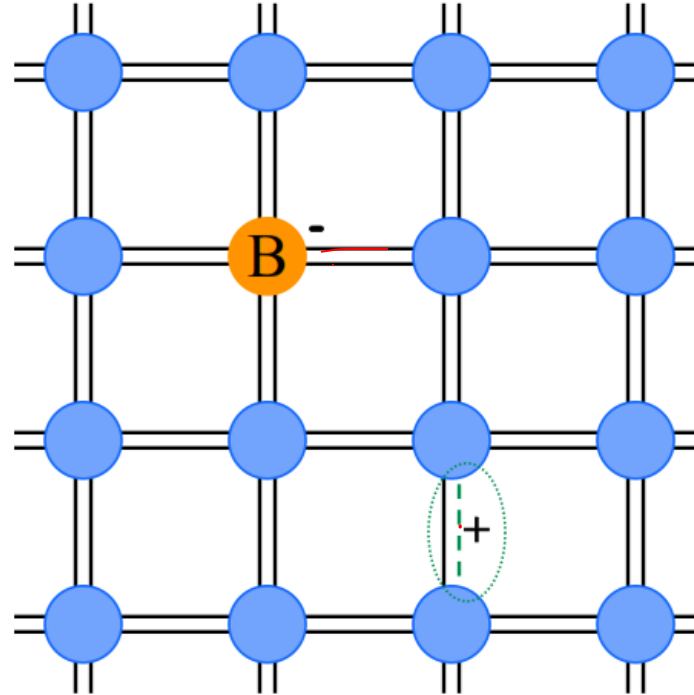
	III	IV	V	VI
	B 5	C 6	N 7	O 8
II	Al 13	Si 14	P 15	S 16
Zn 30	Ga 31	Ge 32	As 33	Se 34
Cd 48	In 49	Sn 50	Sb 51	Te 52
Hg 80	Tl 81	Pb 82	Bi 83	Po 84



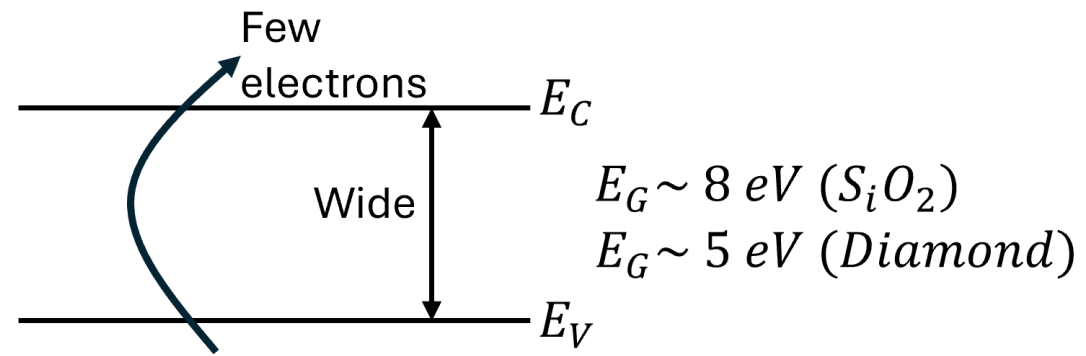
► Extrinsic Semiconductor p-type

III)

	III	IV	V	VI
II	B 5	C 6	N 7	O 8
	Al 13	Si 14	P 15	S 16
Zn 30	Ga 31	Ge 32	As 33	Se 34
Cd 48	In 49	Sn 50	Sb 51	Te 52
Hg 80	Tl 81	Pb 82	Bi 83	Po 84

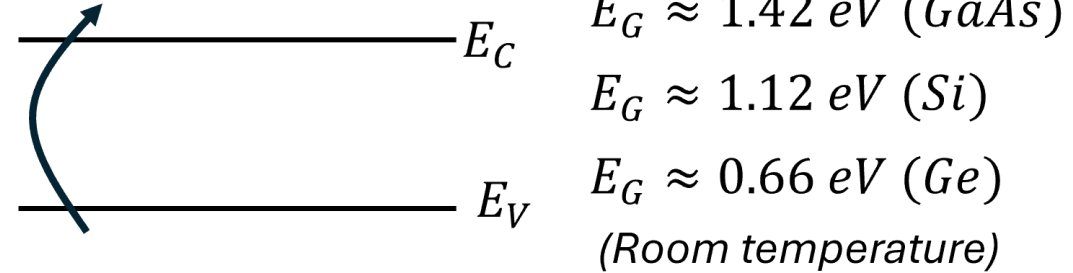


► Material Classification

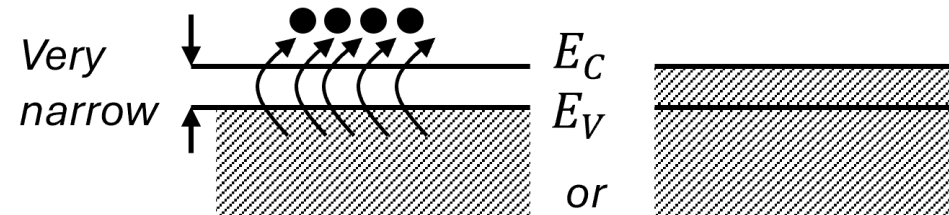


(a) Insulator

Thermal excitation
moderately easy

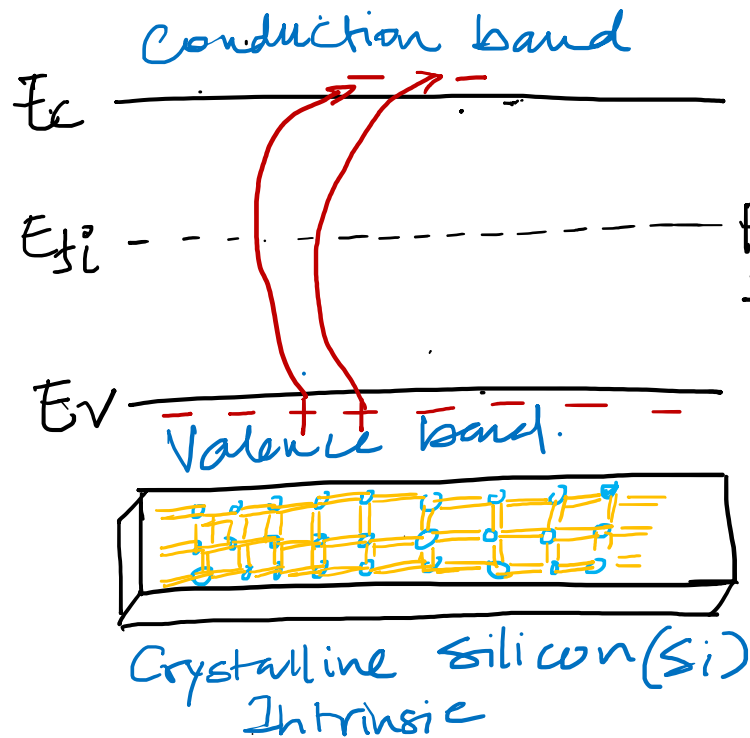


(b) Semiconductor



(c) Metal

► Semiconductor: Intrinsic



Intrinsic carrier conc. @ $T=25^\circ\text{C}$: $n_i = 1.45 \times 10^{10} / \text{cm}^3$

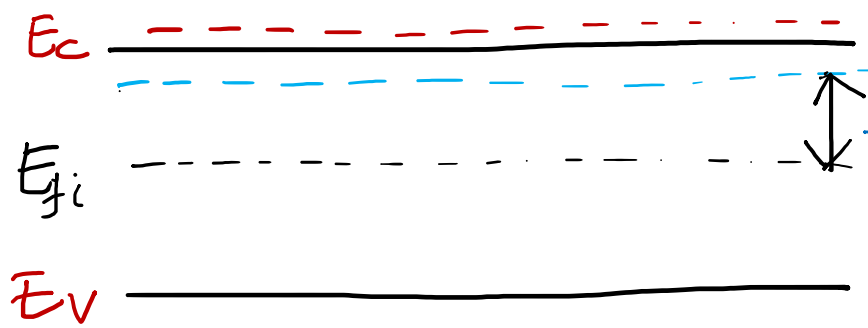
silicon atom density: $N_{Si} = 5 \times 10^{22} / \text{cm}^3$

~ 1 carrier in 10^{12} atoms

electron conc. = $n = n_i$
hole conc. = $p = n_i$

$$n \cdot p = n_i^2 \quad \text{Mass-action Law}$$

► Semiconductor: Extrinsic (n-type)



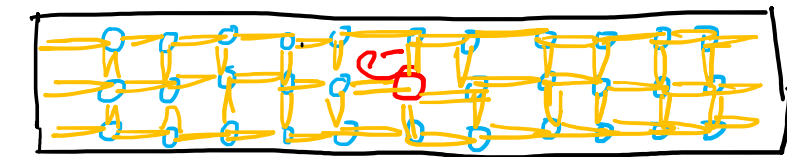
Fermi Potential $\Phi_{Fn} = E_{fi} - E_{dn}$

$$= \frac{KT}{q} \ln \left(\frac{N_D}{N_A} \right)$$

$\underbrace{\frac{KT}{q}}_{V_T \approx 25\text{mV}} \quad \underbrace{\ln \left(\frac{N_D}{N_A} \right)}_{\text{Donor Conc.}}$

$\Phi_{Fn} = -0.45\text{V}$

$\Phi_{Fn} = -0.28\text{V}$



Doped / Extrinsic Silicon

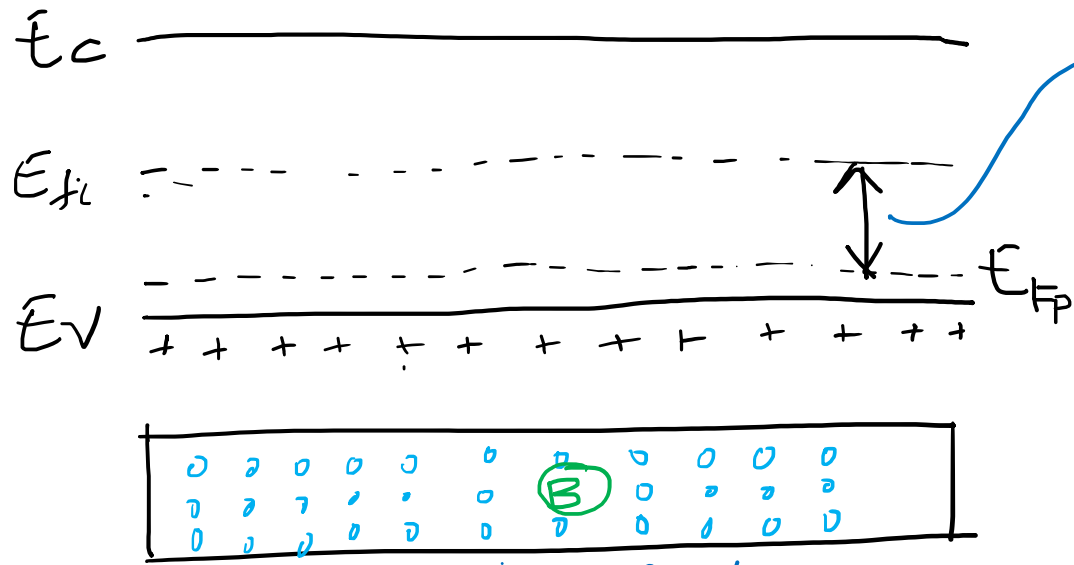
of Phosphorous

$N_{Si} (10^{22}/\text{cm}^3)$ $\gg$ N_D $\gg$ N_A
 $(10^{18}/\text{cm}^3)$ $\dots$ $(10^{15}/\text{cm}^3)$ $(10^{10}/\text{cm}^3)$
 degenerate $\leftarrow$ $n+$ $n, n-$ $\rightarrow$ resistive

electron conc. $n \approx N_D = \underline{\underline{10^{18}}}$ (say)

hole conc. $p = \frac{n_i^2}{n} = \frac{(1.45 \times 10^{10})^2}{10^{18}} \approx \underline{\underline{200}}$

► Semiconductor: Extrinsic (p-type)



Doped semiconductor
of Boron

$$\phi_{FP} = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right)$$

$$N_{Si} \gg N_A \gg n_i$$

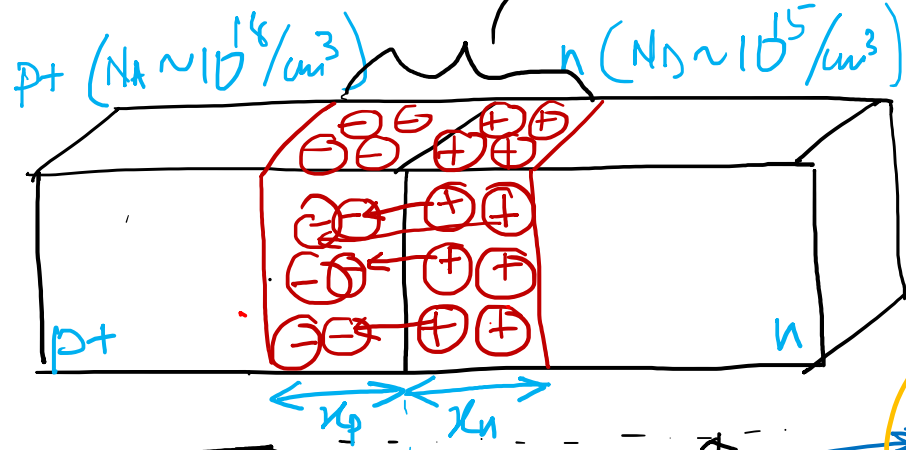
$(\sim 10^{22}/\text{cm}^3)$ $[10^{18}/\text{cm}^3 - 10^{15}/\text{cm}^3]$ $(\sim 10^{10}/\text{cm}^3)$
 degenerate ← → resistive

hole conc. $p \approx N_A = 10^{15}$ (say)

electron conc. $n = \frac{n_i^2}{p} \approx \underline{\underline{2 \times 10^5}}$

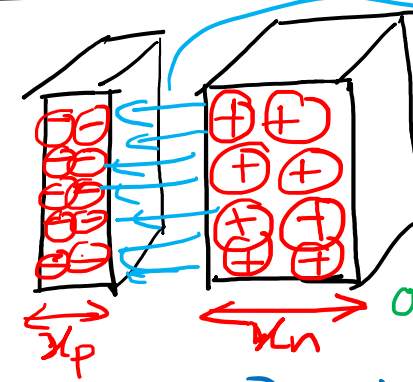
pn-junction

Depletion Charge Q_B



$$Q_{Bn} = q N_D x_n$$

$$= \sqrt{2 q \epsilon_s N_D \phi_0}$$



Gauss's law:

$$E_{peak} = -\frac{Q_p}{\epsilon_s} = \frac{Q_n}{\epsilon_s}$$

or, $\frac{q N_D x_n}{\epsilon_s} = \frac{q N_A x_p}{\epsilon_s}$

$-Q_p = Q_n \Rightarrow -q N_A x_p = q N_D x_n$

or $N_A x_p = N_D x_n$

From Poisson's equation $\Rightarrow d^2V/dn = -\rho/\epsilon$

$$\phi_{op} = \frac{E_{peak} x_p}{2}$$

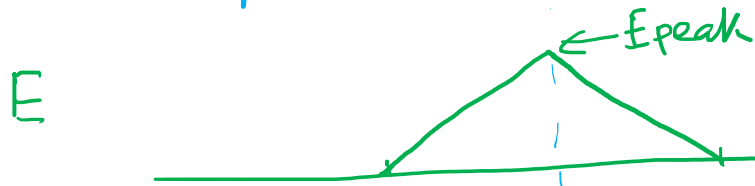
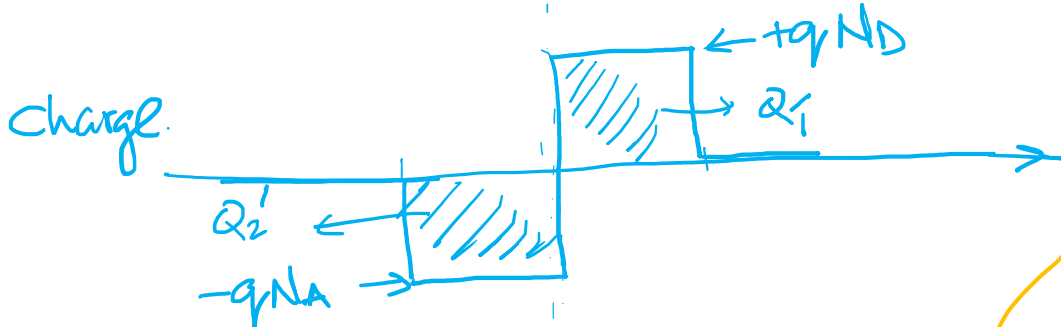
$$\phi_{on} = \frac{E_{peak} x_n}{2}$$

$$\phi_p + \phi_{on} = \phi_{bi} = \frac{q}{2 \epsilon_s} (N_A x_p^2 + N_D x_n^2)$$

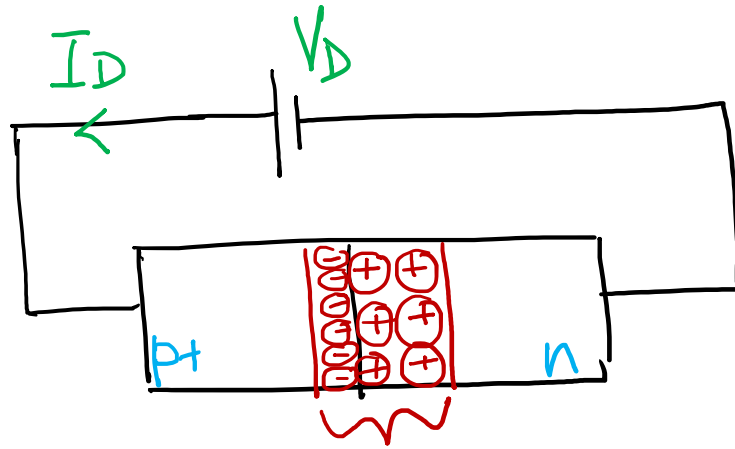
Using $N_A x_p = N_D x_n$ $\therefore x_p = \sqrt{\frac{2 \epsilon_s \phi_{bi}}{q N_A (1 + N_A/N_D)}}$, $x_n = \sqrt{\frac{2 \epsilon_s \phi_{bi}}{q N_D (1 + N_D/N_A)}}$

For one-sided junction

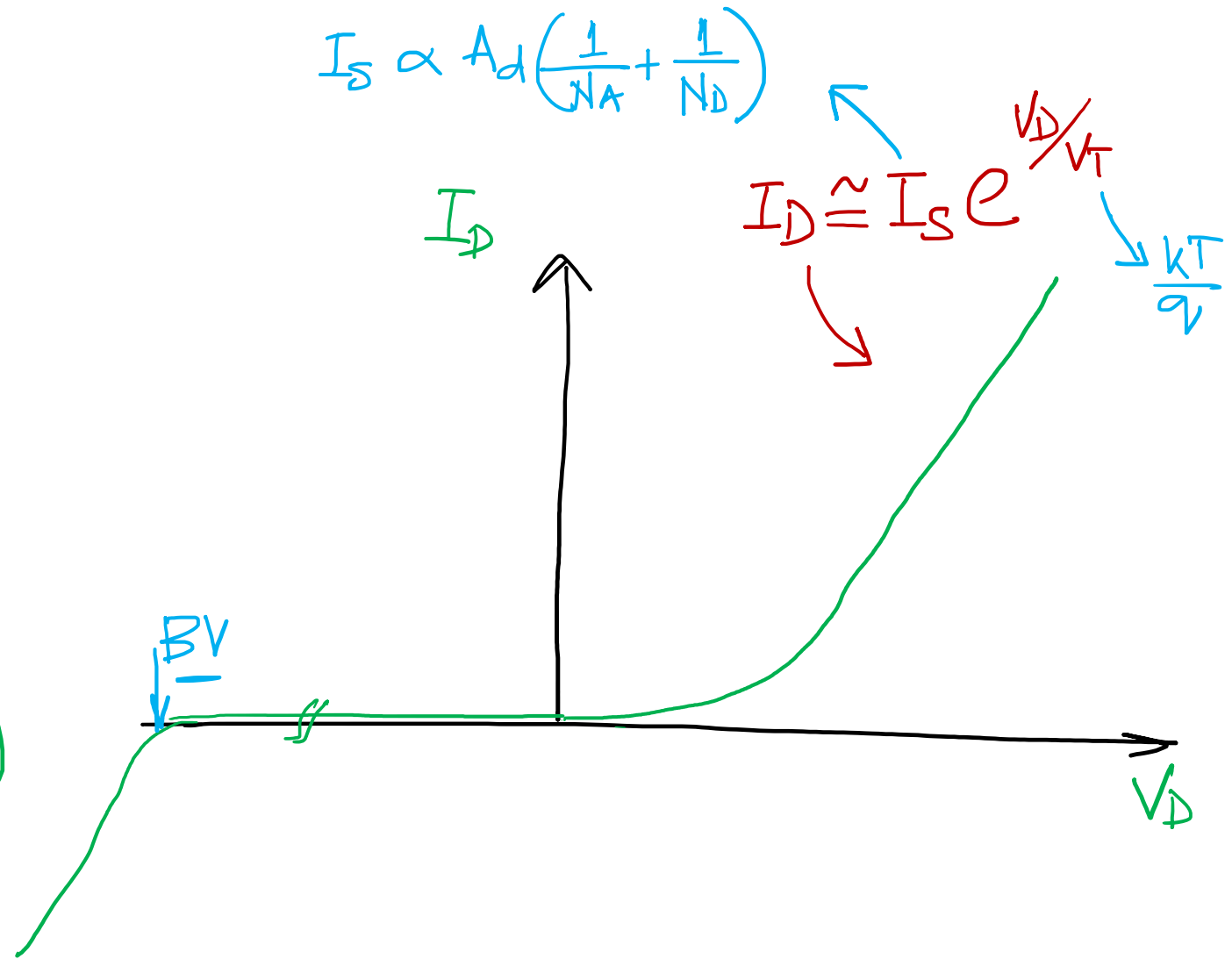
(ie. $N_A \gg N_D$ or $N_D \gg N_A$) $x_p = \sqrt{2 \epsilon_s \phi_{bi} / q N_A}$, $x_n = \sqrt{2 \epsilon_s \phi_{bi} / q N_D}$



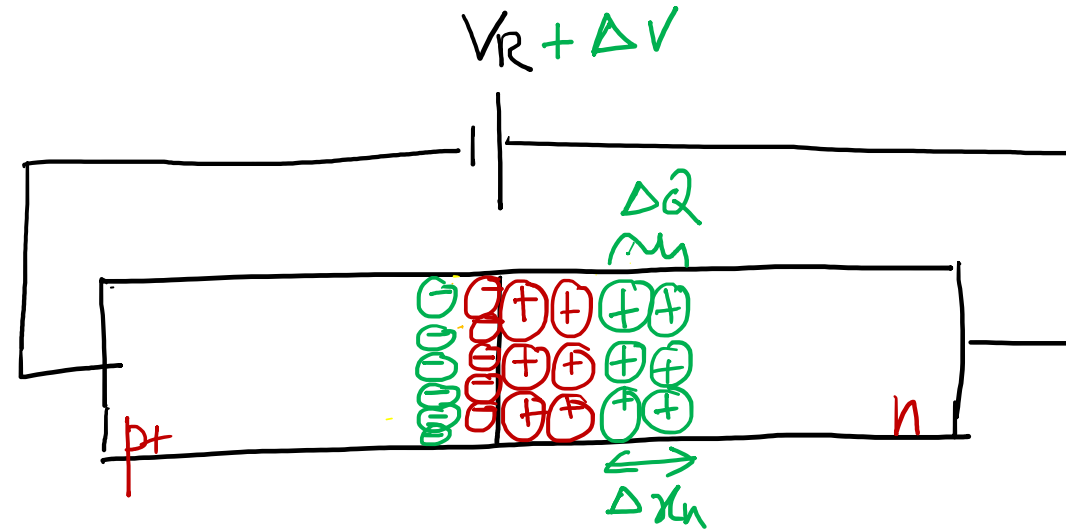
► Forward-biased pn-junction



$$\phi_{bi} = \phi_0 = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$



► Reverse-biased pn-junction



$$C_j = \frac{\Delta Q}{\Delta V} = \frac{\Delta Q}{\Delta x_n} \cdot \frac{\Delta x_n}{\Delta V} \quad \text{or in diff form} \quad C_j = \frac{dQ}{dx_n} \cdot \frac{dx_n}{dV}$$

$$dQ = A \cdot q \cdot N_D \cdot dx_n \quad \& \quad \frac{d}{dV_R}(x_n) = \sqrt{\frac{\epsilon_s}{2qN_D \left(1 + \frac{N_D}{N_A}\right) (\phi_0 + V_R)}}$$

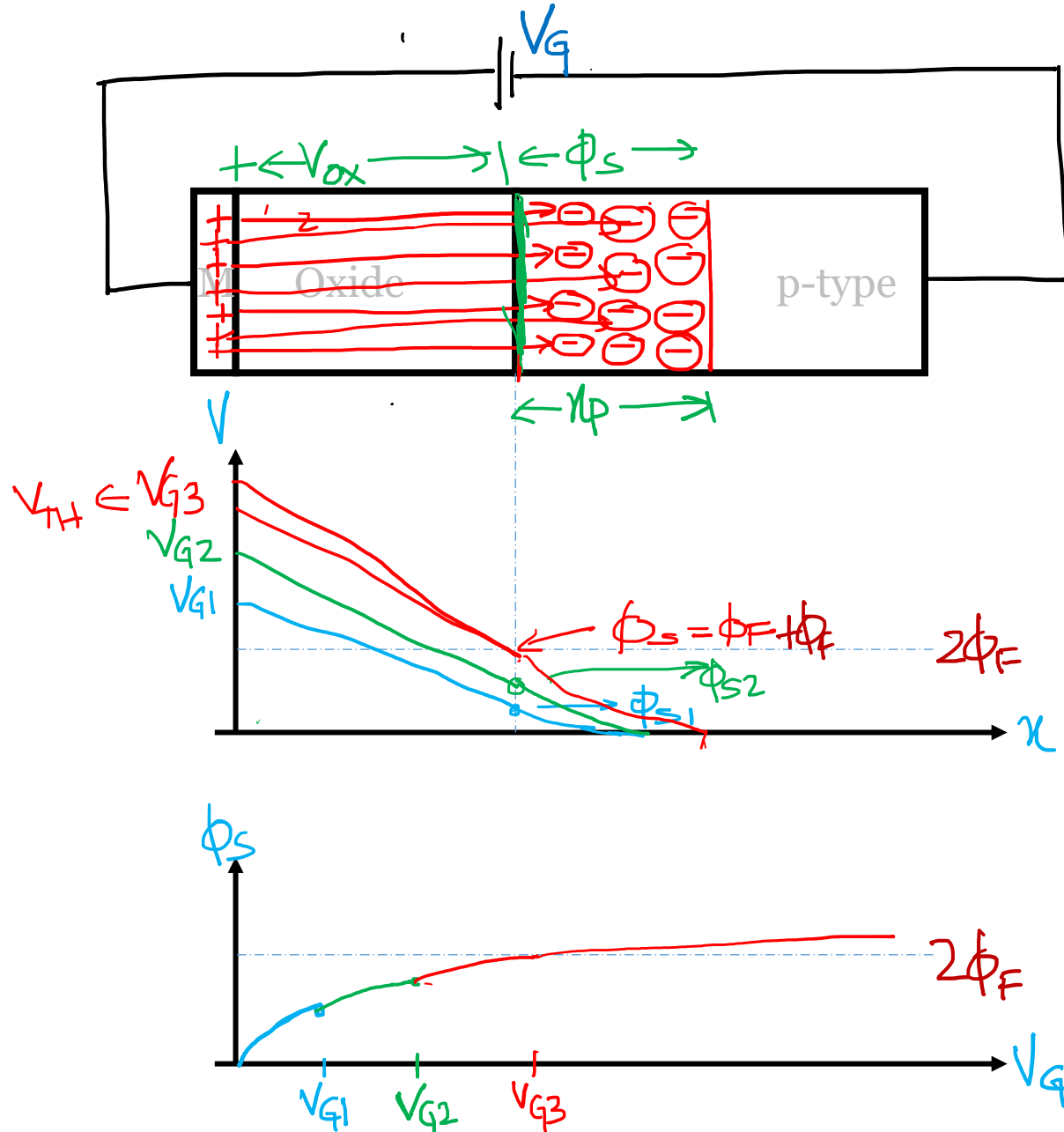
$$\therefore C_j = \overbrace{A \sqrt{\frac{q\epsilon_s N_A N_D}{2(N_A + N_D)}}}^{C_{j0}} \cdot \frac{1}{\sqrt{\phi_0 + V_R}} \quad \text{or}$$

$$C_j = \frac{C_{j0}}{\sqrt{\phi_0 + V_R}}$$

MOS Devices-I

Threshold Voltage

► Metal-Oxide-Semiconductor (MOS): Two-Terminal Device



Applying KVL:

$$V_G = V_{ox} + \phi_s$$

$$V_G = V_{TH} \text{ when } \phi_s = 2\phi_F$$

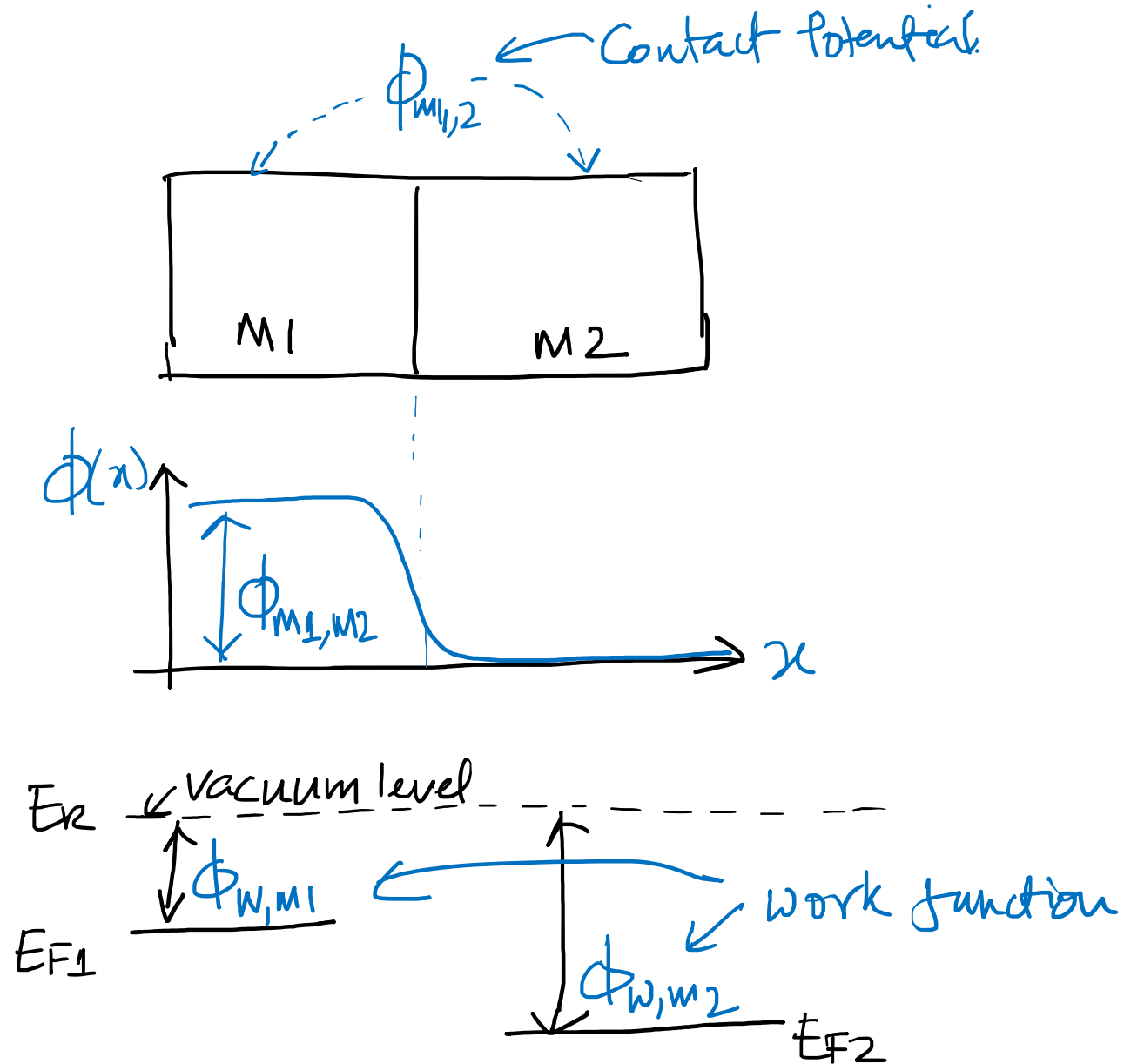
channel is inverted from $p \rightarrow n$

$$V_{TH} = V_{ox} + 2|\phi_F|$$

$$= \frac{Q_{dep}}{C_{ox}} + 2|\phi_F|$$

$$V_{TH} \text{ (ideal)} = \frac{\sqrt{2\epsilon_s N_A 2|\phi_F|}}{C_{ox}} + 2|\phi_F|$$

► Contact Potential



A vertical stack of metal layers is shown, labeled $M_1, M_2, \dots, M_{n-1}, M_n$. Dashed arrows connect the top of M_1 and the bottom of M_n to the equations below.

$$\begin{aligned}\phi_{\text{contact}} &= \phi_{w,m1} - \phi_{w,mn} \\ &= \underline{\phi_{m1 - mn}}\end{aligned}$$

► Metal-Oxide-Semiconductor (MOS): Two-Terminal Device

Contact Potential



$\phi_G - \phi_S$
 $\phi_{GS} = V_{FB} \leftarrow \text{Flatband voltage}$

Example: For a n-polysilicon gate with doping conc. $N_{d,poly}$

$$\phi_{GS} = V_{FB} = -\frac{KT}{q} \ln \left(\frac{N_A \cdot N_{d,poly}}{n_i^2} \right)$$

$$V_{TN} = V_{FB} + \frac{\sqrt{2qN_A^2|\phi_F|}}{C_{ox}} + z|\phi_F|$$

Example: $C_{ox} = 3.45 \text{ fF}/\mu\text{m}^2$
 $N_A = 10^{15} / \text{cm}^3$, $N_{d,poly} = 10^{19} / \text{cm}^3$

$$V_{FB} = -0.865 \text{ V}, \quad z|\phi_F| = 0.579 \text{ V}$$

$$\frac{Q_{dep}}{C_{ox}} = \frac{Q_B}{C_{ox}} \approx 0.04 \text{ V}$$

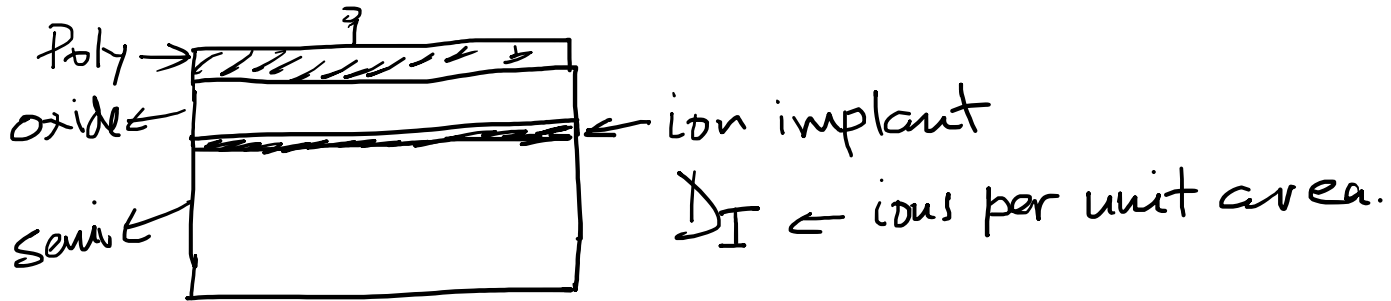
$$\Rightarrow V_{TN} \approx -0.865 + 0.04 + 0.579$$

or

$$V_{TN} \approx -0.25 \text{ V}$$

depletion mode device

Enhancement Mode Device $V_{tn} > 0V$



$$V_{TN} = V_{FB} + \underbrace{\frac{Q_B}{C_{ox}} + 2|\phi_F|}_{\text{intrinsic}} + \underbrace{\frac{qD_I}{C_{ox}}}_{\text{controlled implant}}$$

Previous example case:

$$D_I = 2 \times 10^{12} / \text{cm}^2$$

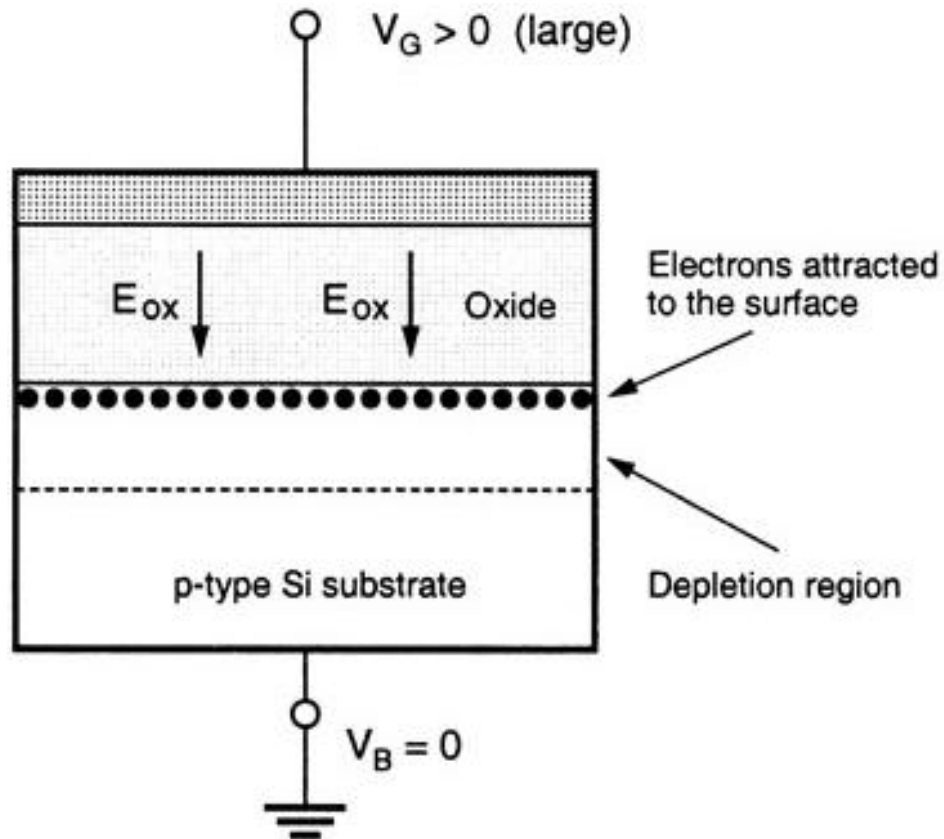
$$\Rightarrow \frac{qD_I}{C_{ox}} = \underline{0.928V}$$

$$\begin{aligned} V_{TN_{en}} &= -0.25 + 0.928 \\ &= 0.678V \end{aligned}$$

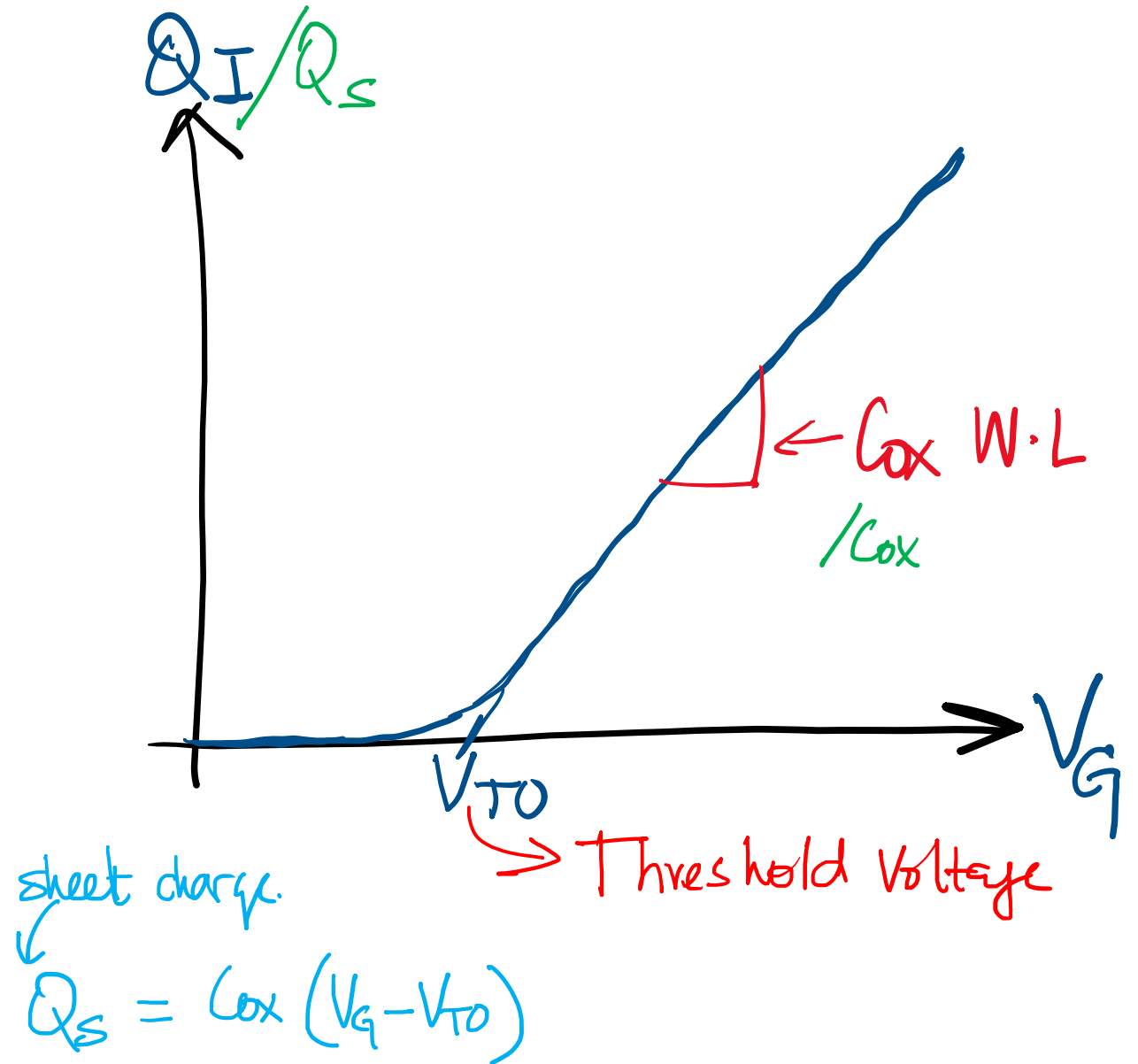
Complete V_t Equation:

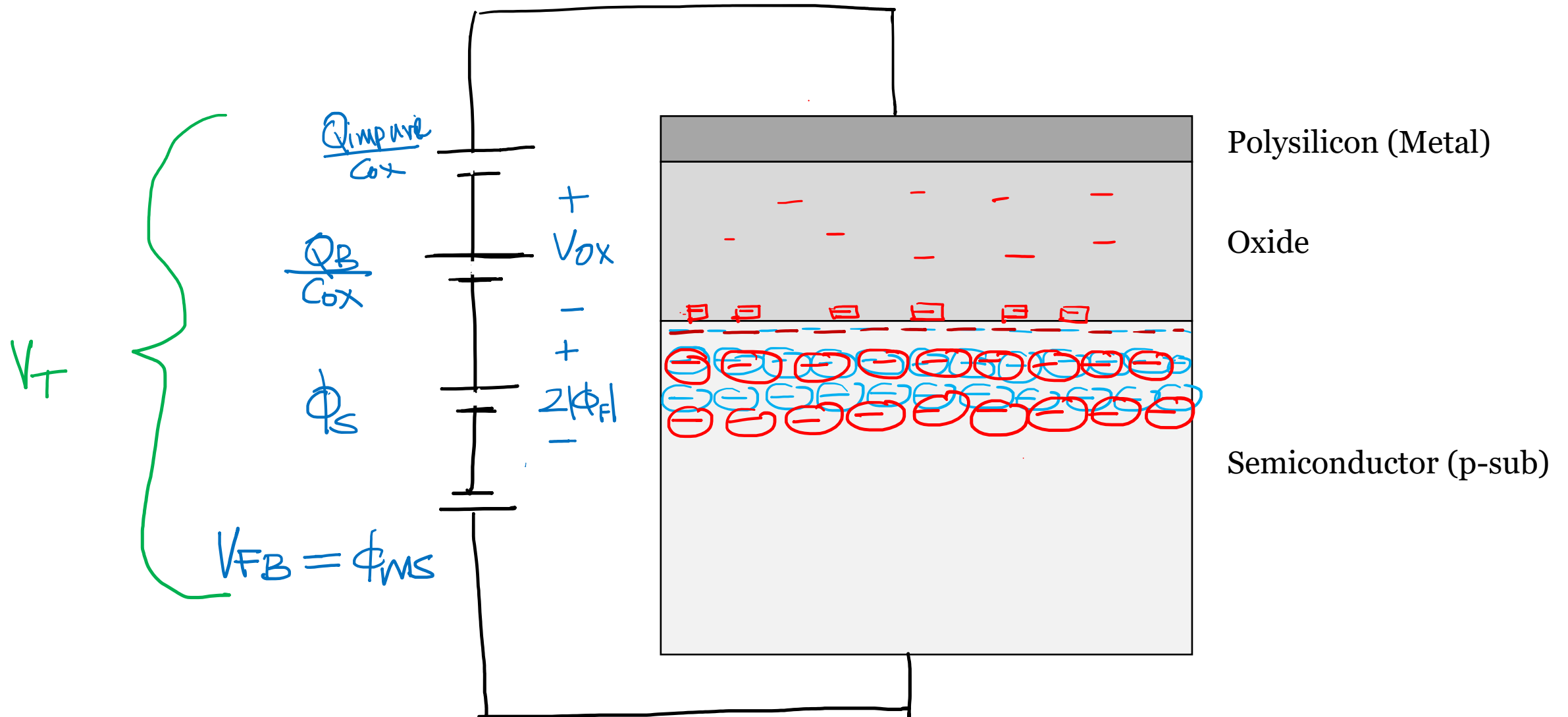
$$V_{TN} = V_{FB} + \frac{Q_B}{C_{ox}} + 2|\phi_F| + \frac{qD_I}{C_{ox}} + \frac{Q_{\text{impure}}}{C_{ox}}$$

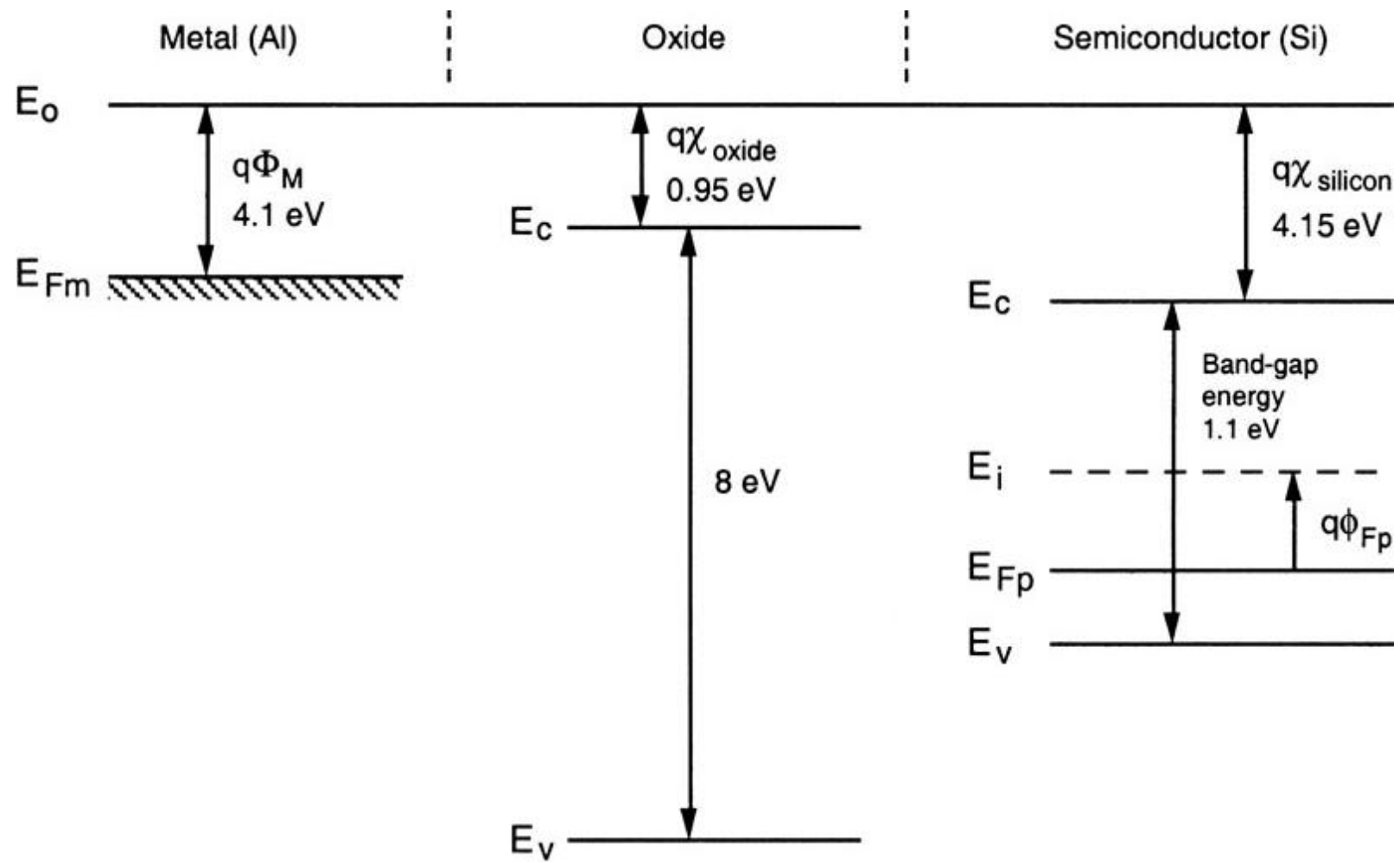
► MOS Capacitance

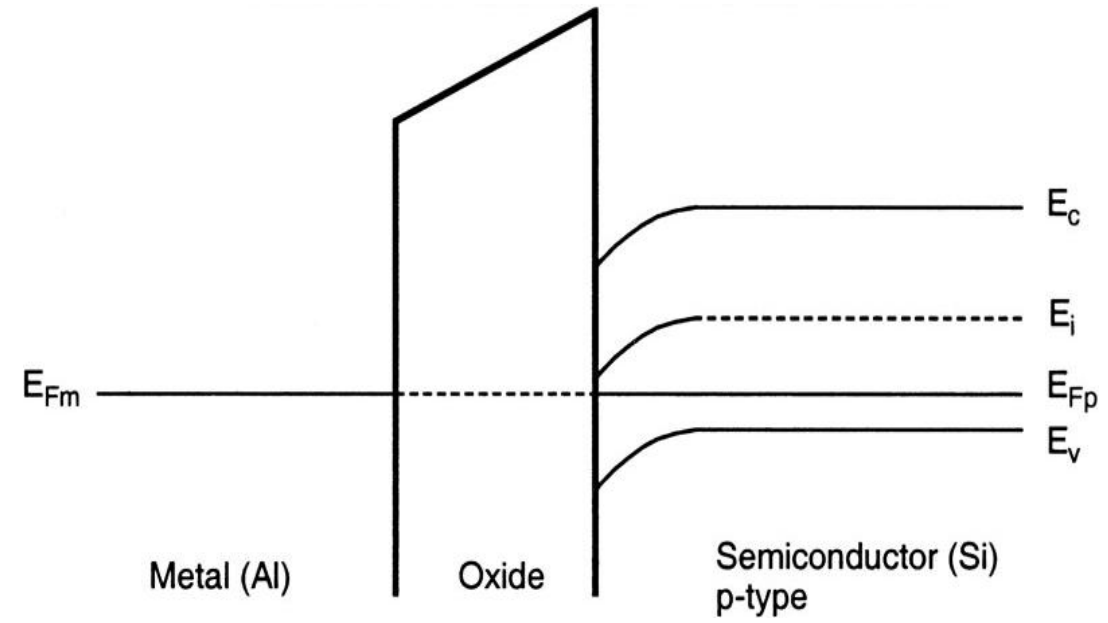
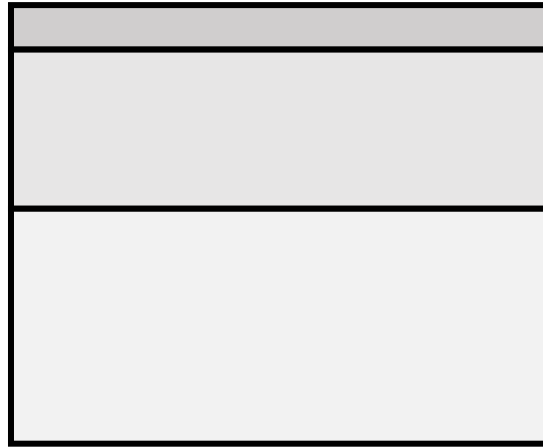


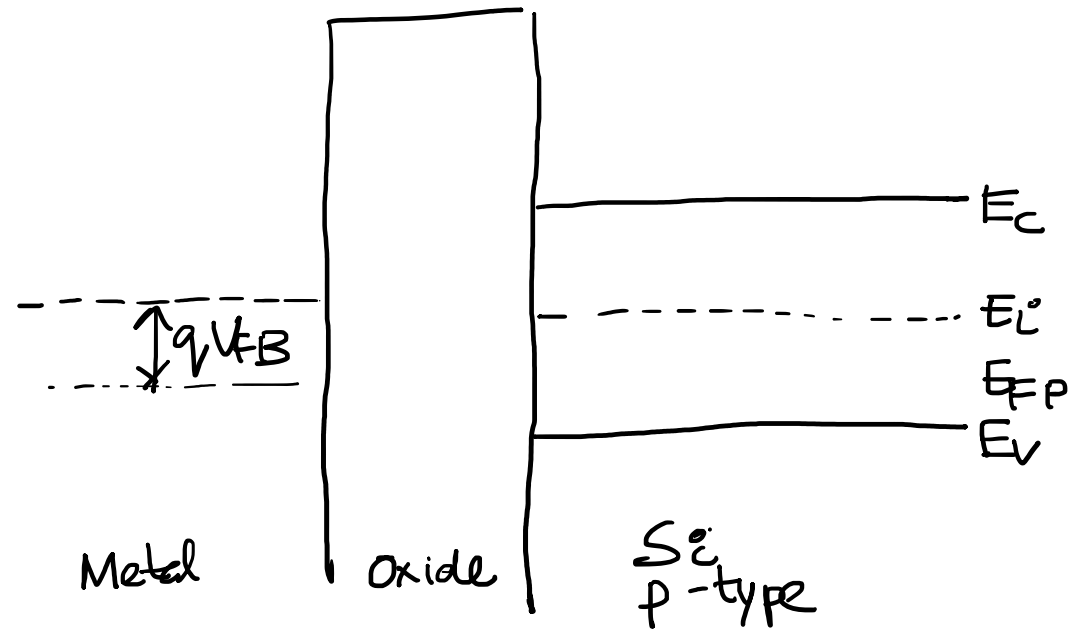
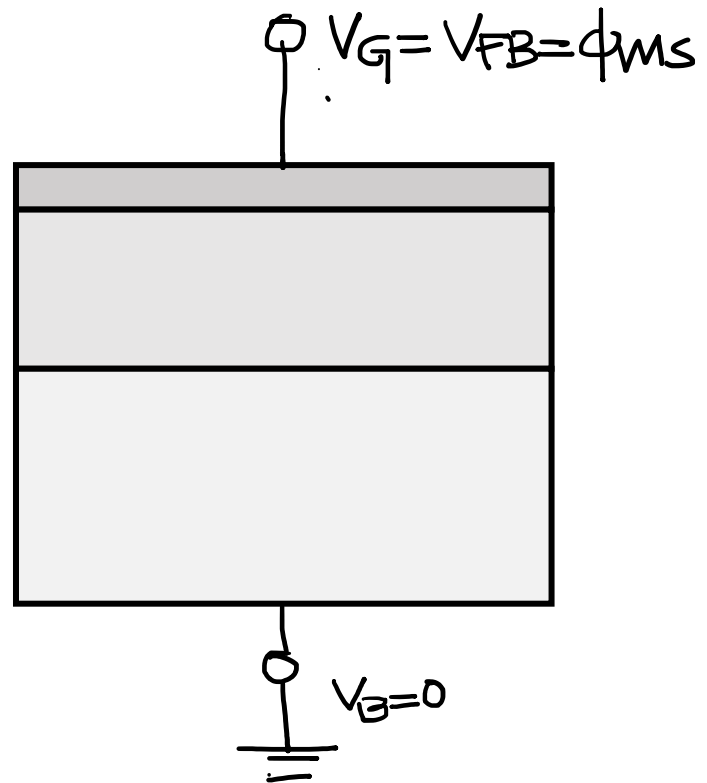
$$Q_I = C_{ox} \cdot W \cdot L \cdot (V_G - V_{TO})$$

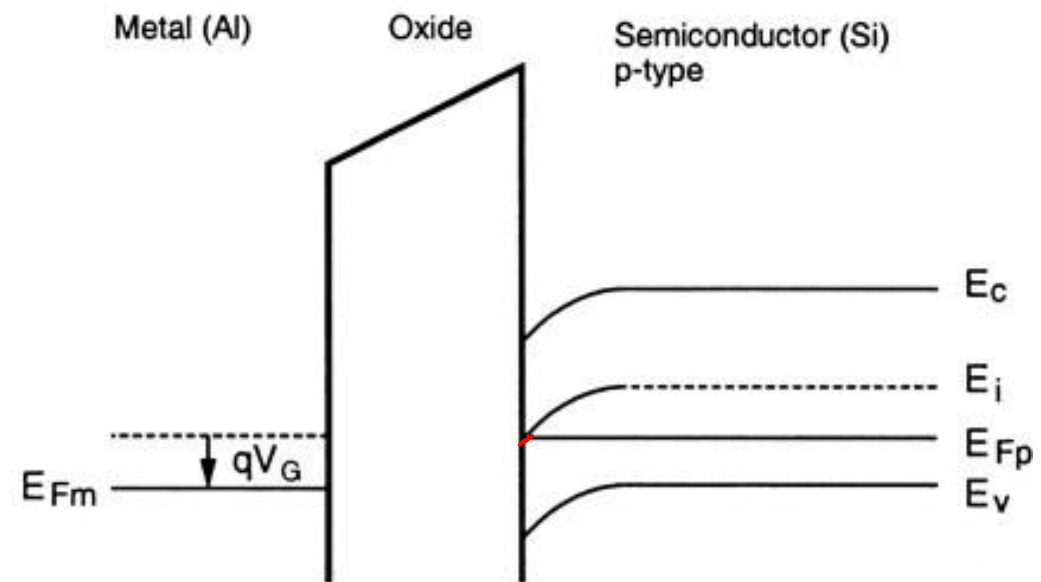
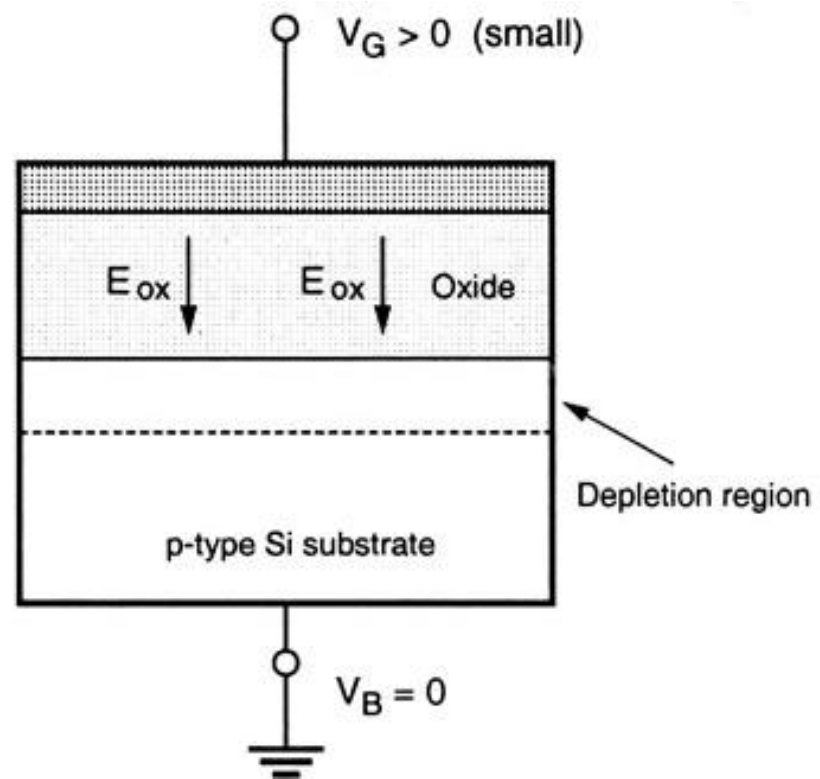


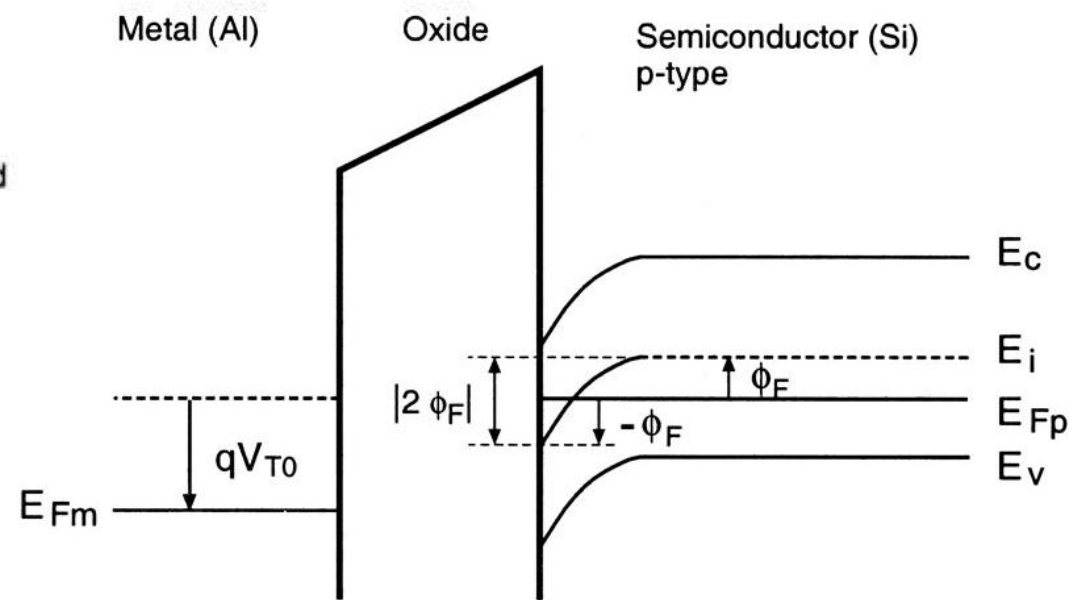
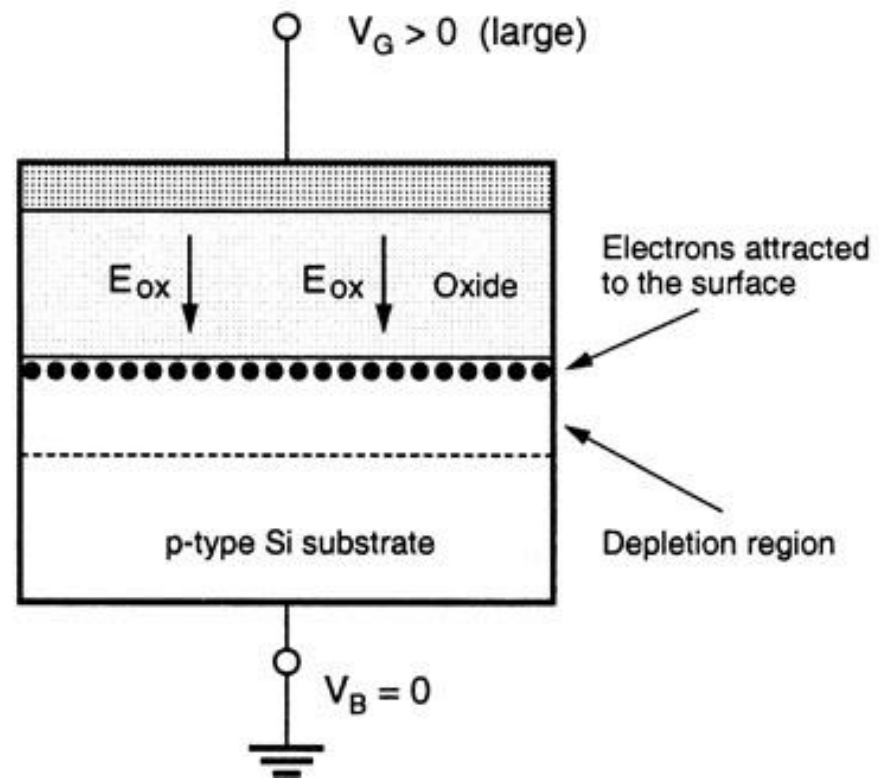




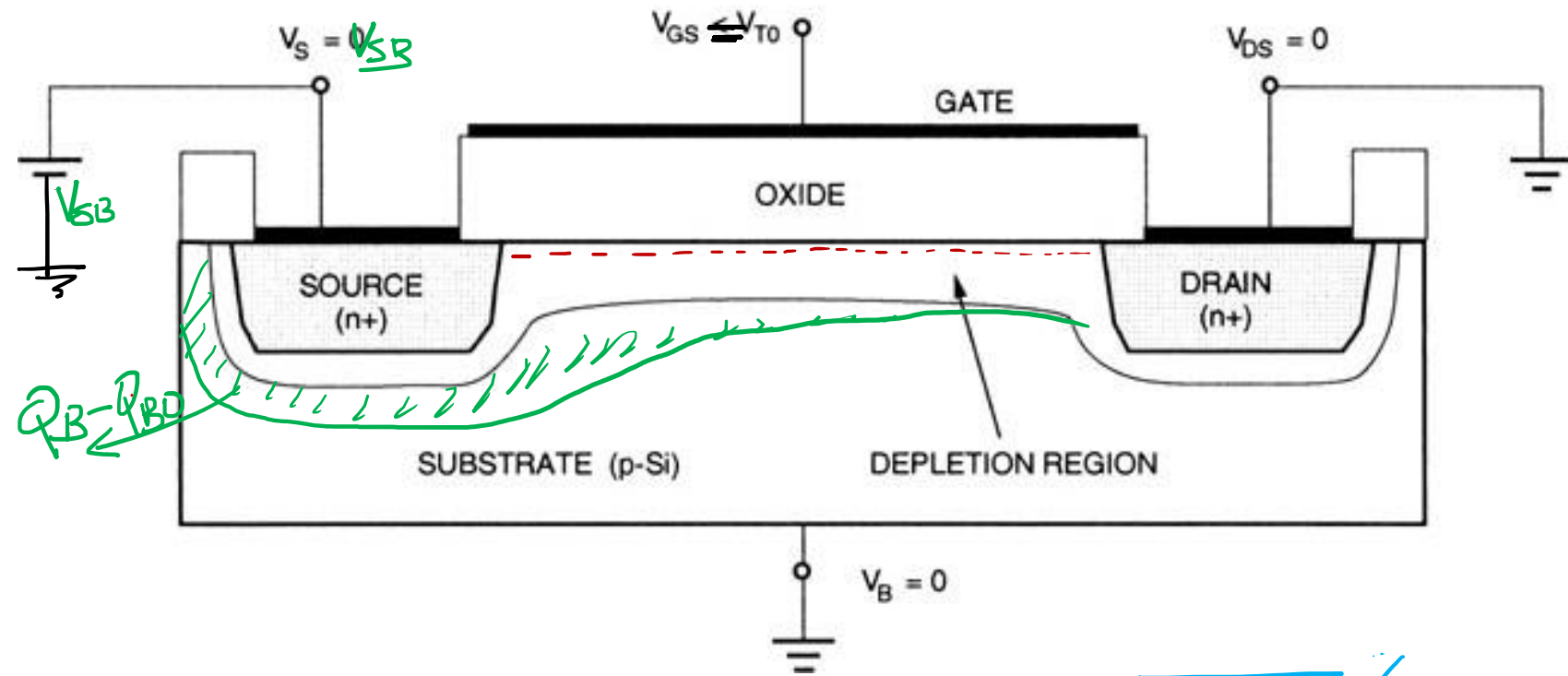








► Body Effect



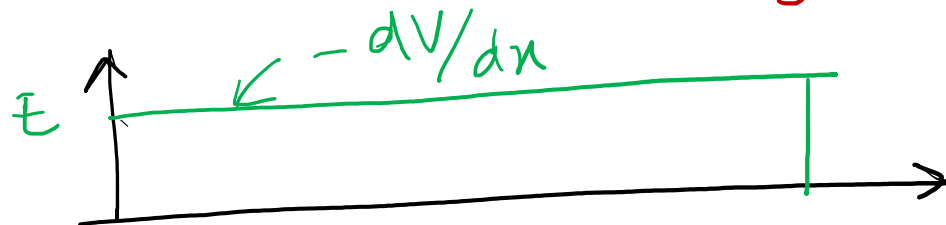
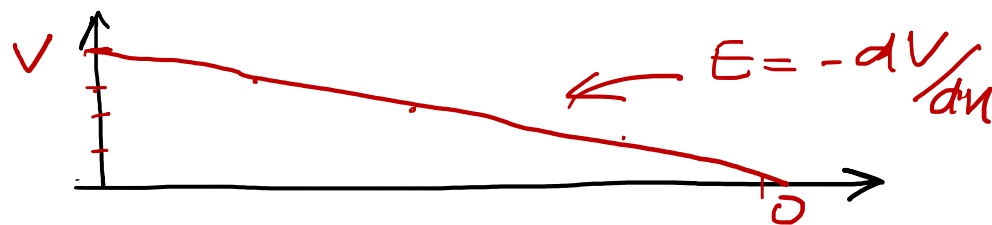
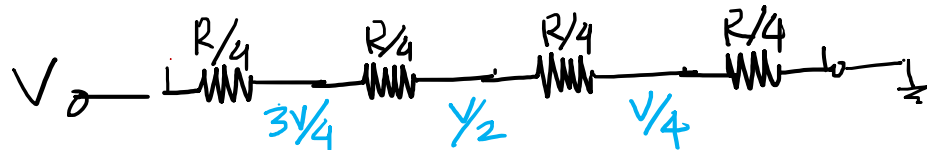
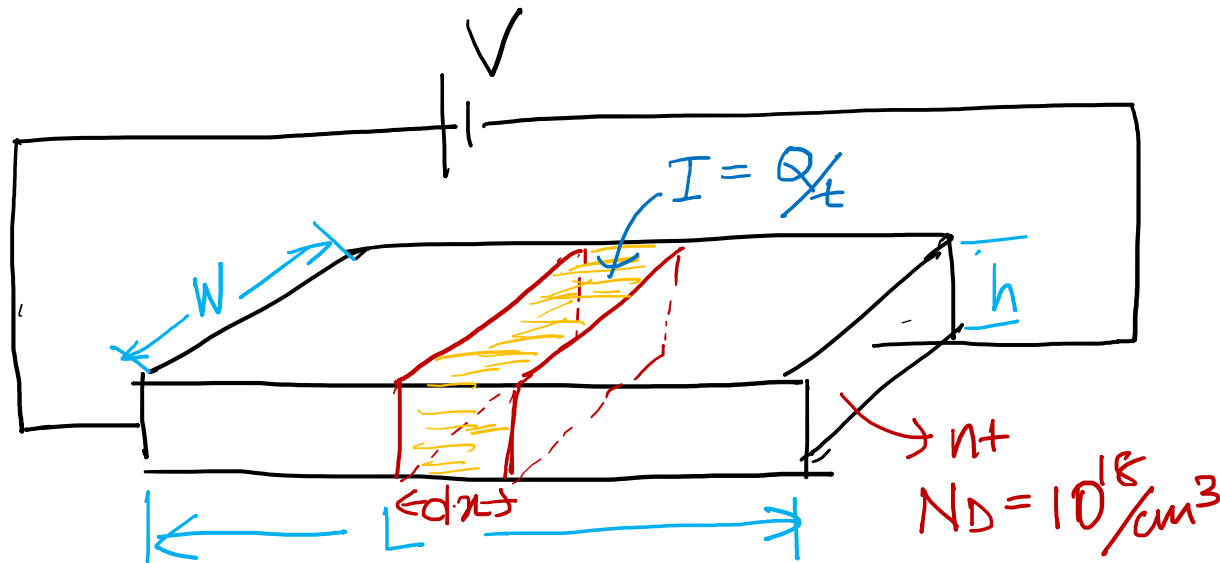
$$V_T(V_{SB}) = V_{T0} + \frac{Q_B - Q_{BD}}{C_{ox}} = V_{T0} + \underbrace{\frac{\sqrt{2q\epsilon_s N_A}}{C_{ox}}}_{\gamma} \left(\sqrt{2|\phi_F| + V_{SB}} - \sqrt{2|\phi_F|} \right)$$

$$V_T = V_{T0} + \gamma \left(\sqrt{2|\phi_F| + V_{SB}} - \sqrt{2|\phi_F|} \right)$$

MOS Device II

I-V Characteristics

► Conduction in Semiconductors : Drift



$$F = qE = ma$$

$$v_d \propto E \Rightarrow v_d = \mu E$$

avg velocity of electrons

mobility of electrons

$$\text{Avg. current} = \underbrace{\frac{\text{Charge}}{dx}}_{Q \text{ per unit } L} \times \underbrace{\frac{dv}{dt}}_{\text{velocity } v_d}$$

$$I = Q_s \cdot W \cdot v_d = Q_s \cdot \frac{W}{L} \mu \cdot V$$

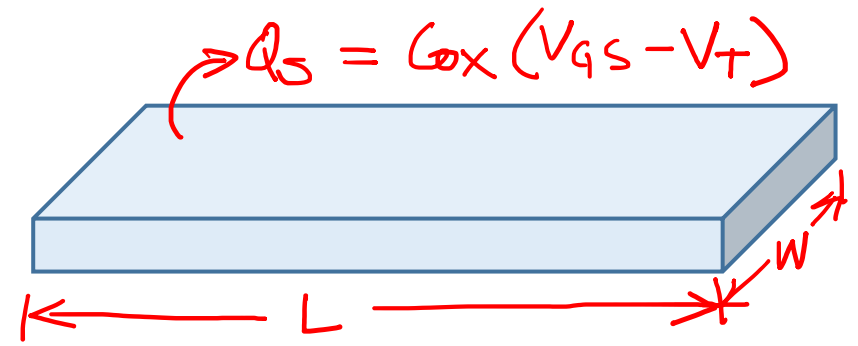
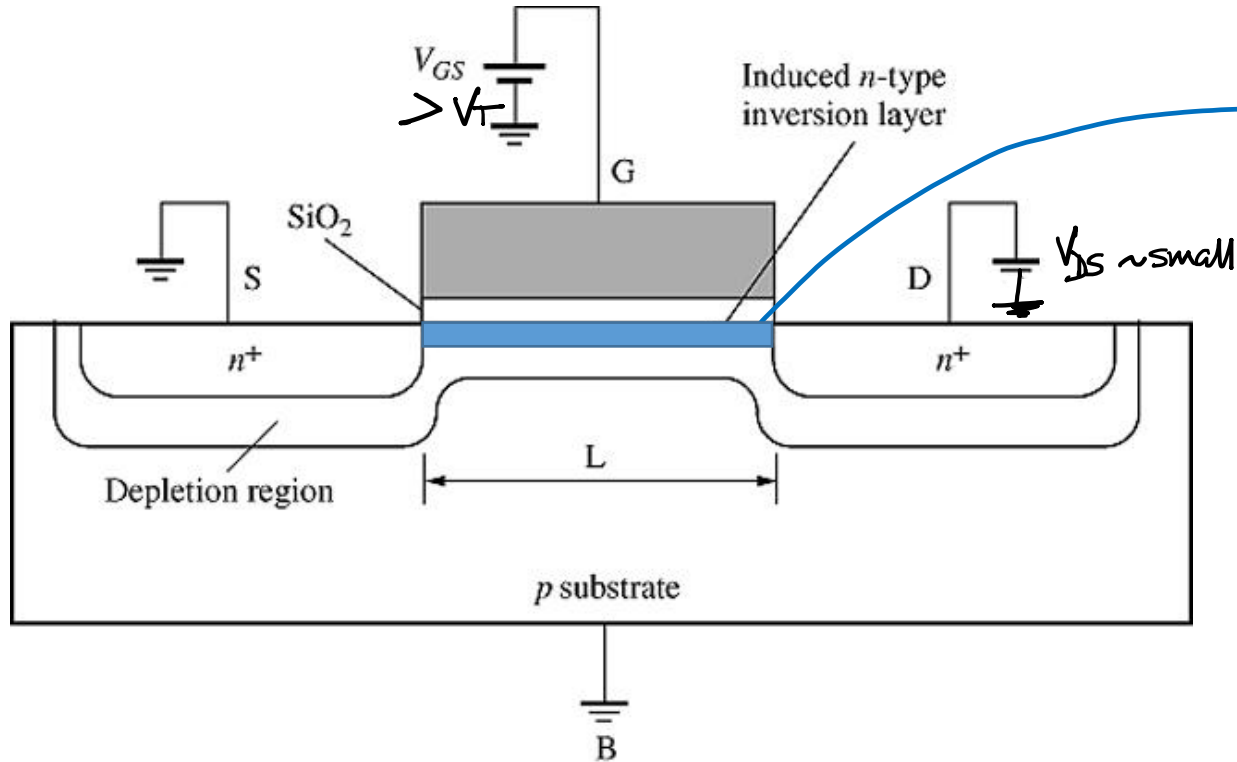
$$\text{sheet charge density } (Q_s) = \frac{N_D \cdot q \cdot W \cdot L \cdot h}{W \cdot L} = N_D q h$$

$$V = \left(\frac{1}{Q_s \mu} \right) \left(\frac{L}{W} \right) \cdot I$$

I_{sheet}

$$R = I_{\text{sheet}} \cdot L/W$$

► I_{DS} for Small V_{DS}

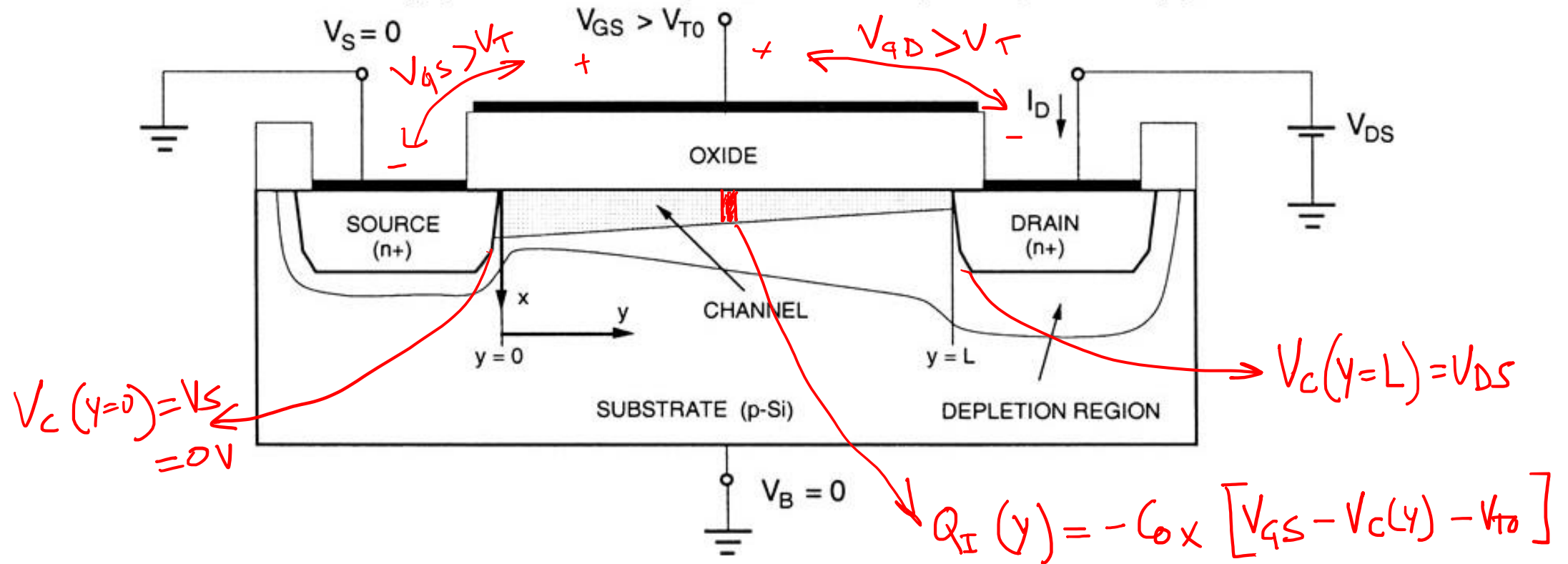


$$I_{DS} = Q_S \cdot \mu_n \cdot \frac{W}{L} \cdot V_{DS}$$

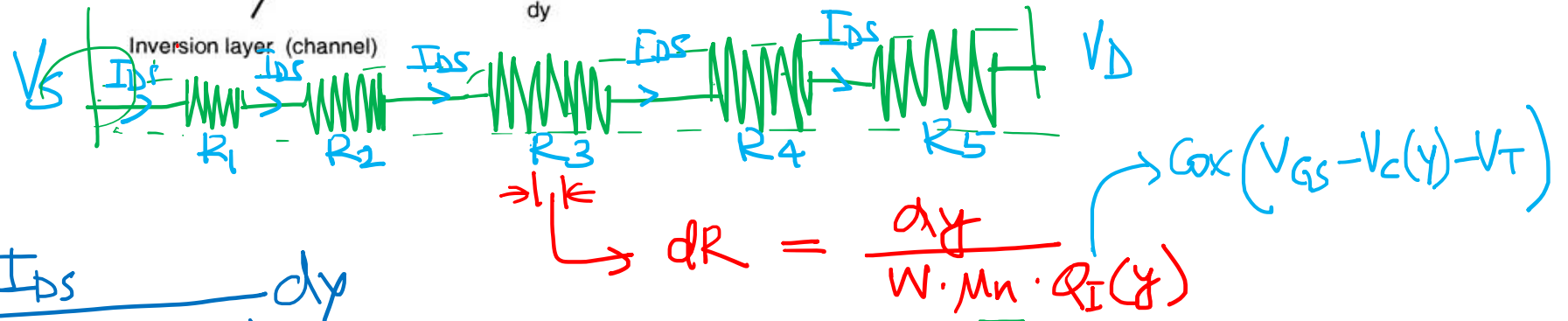
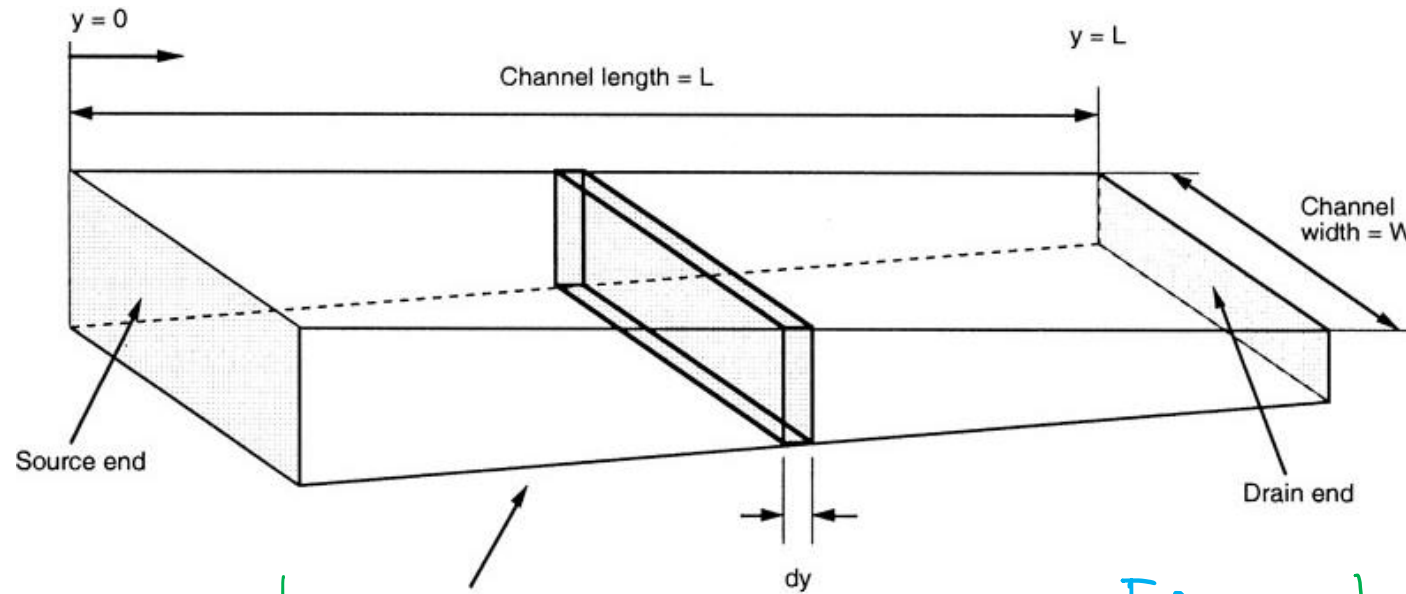
$$I_{DS} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T) V_{DS}$$

$V_{GS} > V_T$
 $V_{DS} \sim \text{small}$ } Voltage-controlled resistor.

► Gradual Channel Approximation



► Gradual Channel Approximation



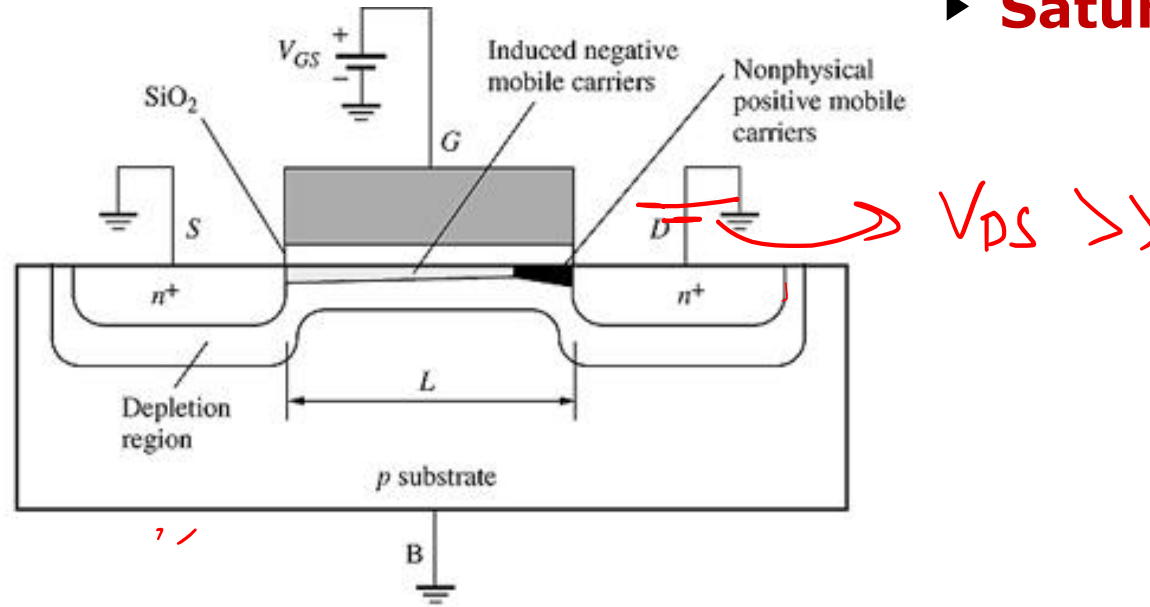
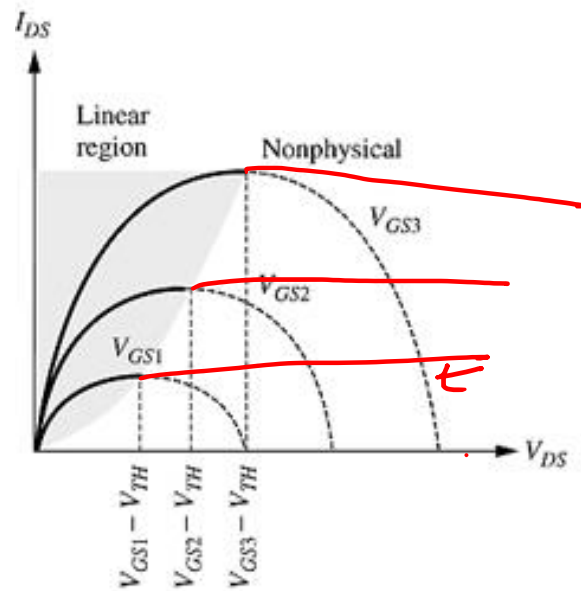
$$dV_c = I_{DS} \cdot dR = \frac{I_{DS}}{W \cdot \mu_n \cdot Q_I(y)} dy$$

$$I_{DS} \int_0^L dy = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_c(y) - V_T) dV_c$$

$$\Rightarrow I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \left[2(V_{GS} - V_T) V_{DS} - V_{DS}^2 \right]$$

$V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$, Linear mode

► Saturation (Pinch-Off) Region



If $I_{DS} = \text{constant}$ for $V_{DS} > V_{GS} - V_T$

$$I_{DS(\text{sat})} = I_{DS(\text{lin})}(V_{DS} = V_{GS} - V_T)$$

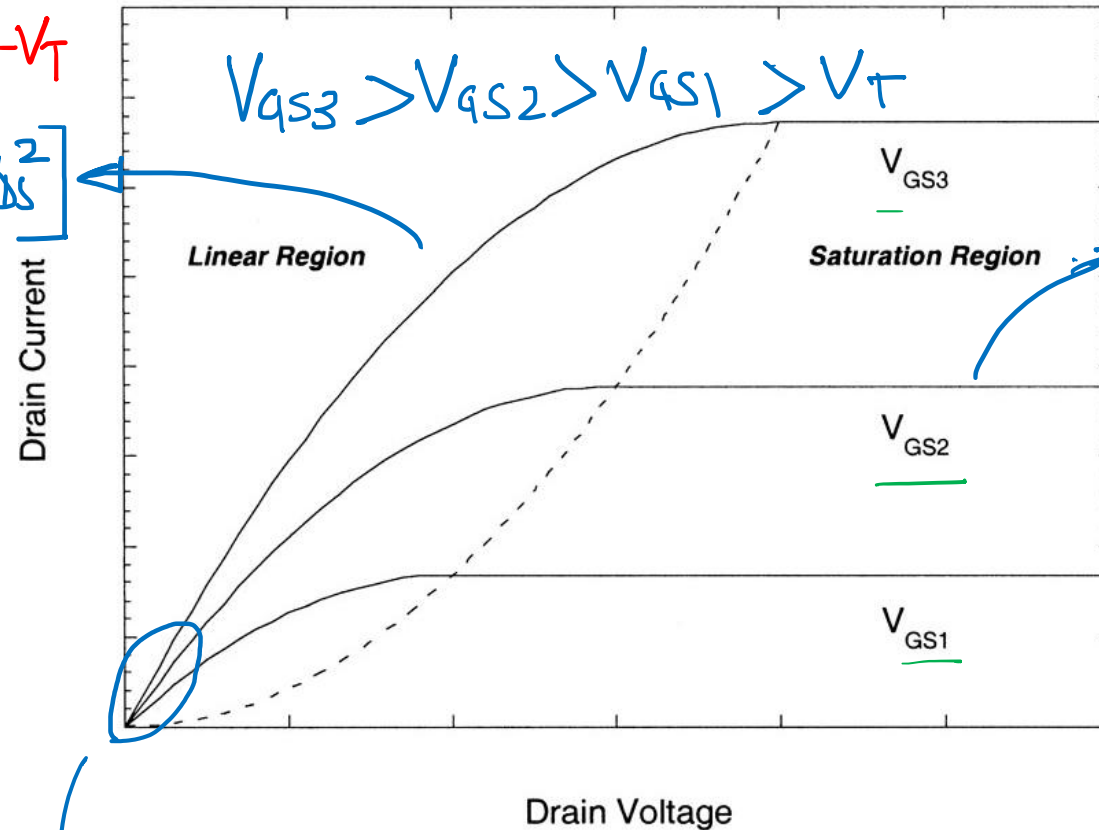
$$I_{DS(\text{sat})} = \frac{\mu_n \epsilon_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2$$

$$\therefore I_{DS(\text{sat})} = \frac{\mu_n \epsilon_{ox}}{2} \frac{W}{L} \left[\underbrace{2(V_{GS} - V_T)}_{V_{DS}} \underbrace{(V_{GS} - V_T)}_{V_{DS}} - \underbrace{(V_{GS} - V_T)^2}_{V_{DS}} \right]$$

► I_{DS} vs. V_{DS} (V_{GS} const.)

For $V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2 \right]$$



For $V_{GS} > V_T$, $V_{DS} \gg V_{GS} - V_T$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2$$

For $V_{GS} > V_T$, $V_{DS} \sim \text{small}$

$$I_{DS} \cong \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T) V_{DS}$$

↳ voltage-controlled (V_{GS}) resistor

► Channel Length Modulation

From gradual channel approx:

$$I_{DS} \int_0^{L'} dy = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_c(y) - V_T) dV_c$$

$\text{L} \rightarrow \text{L}'$ $V_{DS} \rightarrow V_{DSAT}$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \cdot \frac{W}{L'} (V_{GS} - V_T)^2$$

$$L' = L - \Delta L$$

$$\Delta L \propto \sqrt{V_{DS} - V_{DSAT}}$$

$$I_{DS}(sat) = \frac{\mu_n C_{ox}}{2} \cdot \frac{W}{L \left(1 - \frac{\Delta L}{L}\right)} (V_{GS} - V_T)^2$$

$\downarrow$ empirically
 $1 - \lambda V_{DS}$

Assuming $\lambda V_{DS} \ll 1$ i.e. $1/(1 - \lambda V_{DS}) \approx (1 + \lambda V_{DS})$

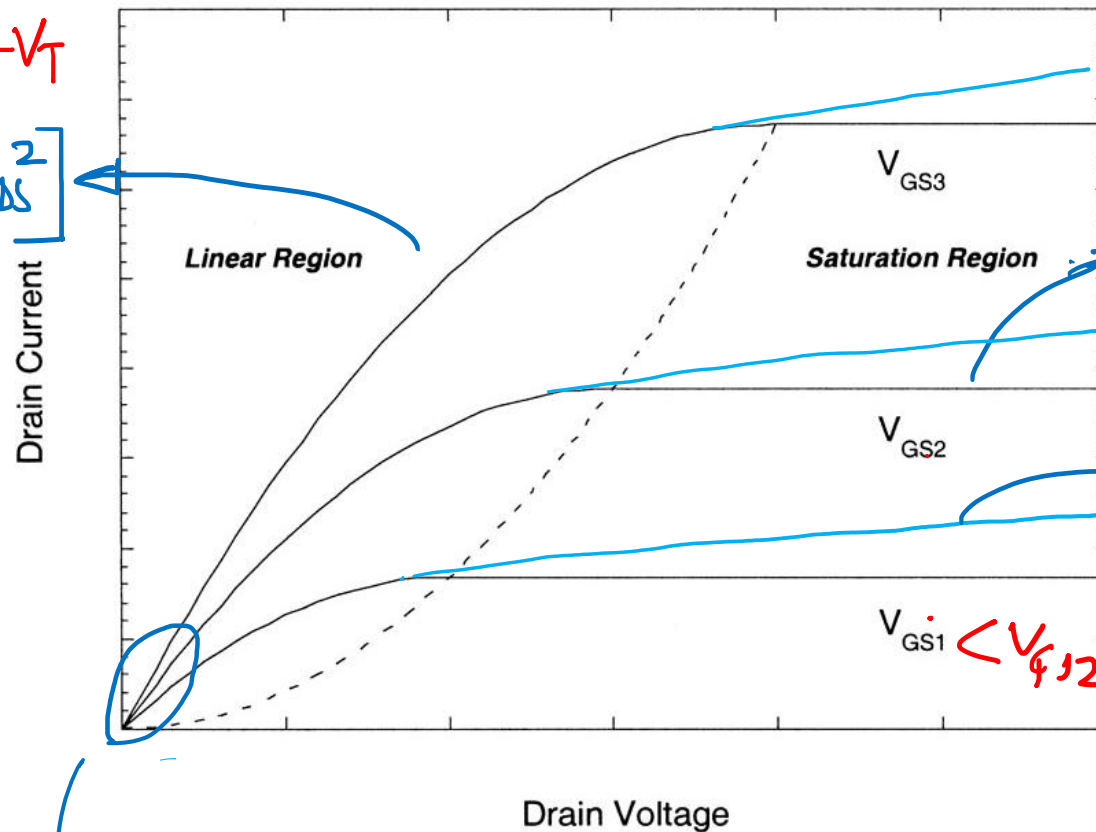
$$I_{DS}(sat) = \frac{\mu_n C_{ox}}{2} \cdot \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$V_{GS3} > V_{GS2} > V_{GS1} > V_T$$

► I_{DS} vs. V_{DS} (V_{GS} const.)

For $V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

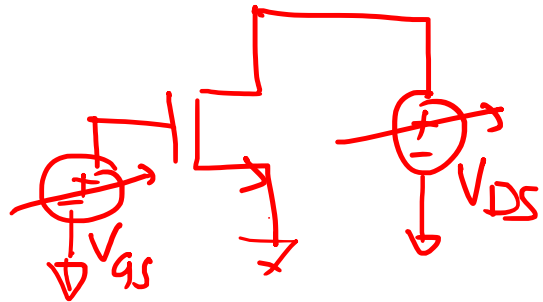


For $V_{GS} > V_T$, $V_{DS} \gg V_{GS} - V_T$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2$$

$$I_{DS} = I_{DSsat} (1 + \lambda V_{DS})$$

$$V_{GS1} < V_{GS2} < V_{GS3}$$



For $V_{GS} > V_T$, $V_{DS} \sim \text{small}$

$$I_{DS} \cong \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T) V_{DS}$$

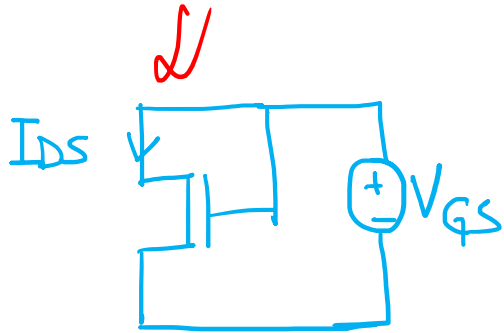
↳ voltage-controlled (V_{GS}) resistor

► I_{DS} vs. V_{GS} ($V_{DS} > V_{GS} - V_T$)

$$V_{DS} > V_{GS} - V_T$$

$$V_{GS} = V_{DS}$$

I_{DS}
Drain Current



V_{T0}

Gate Voltage

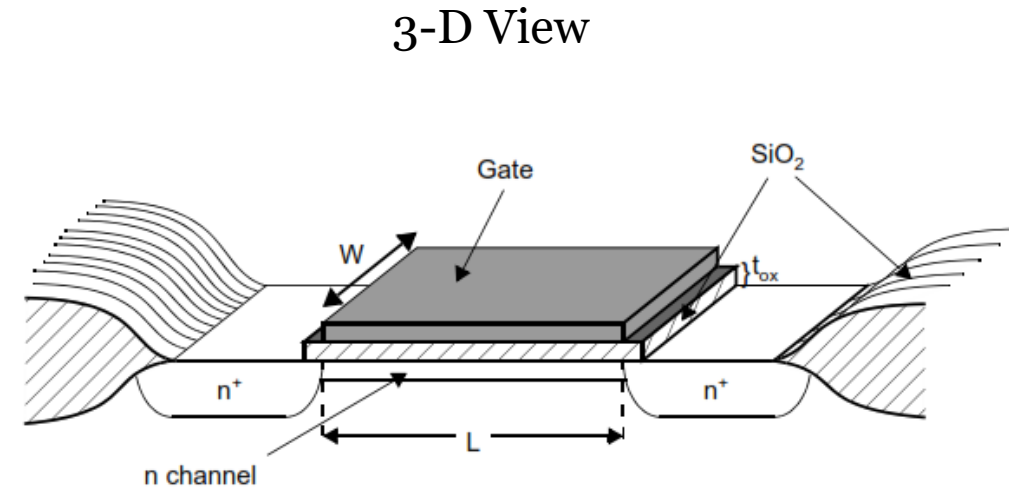
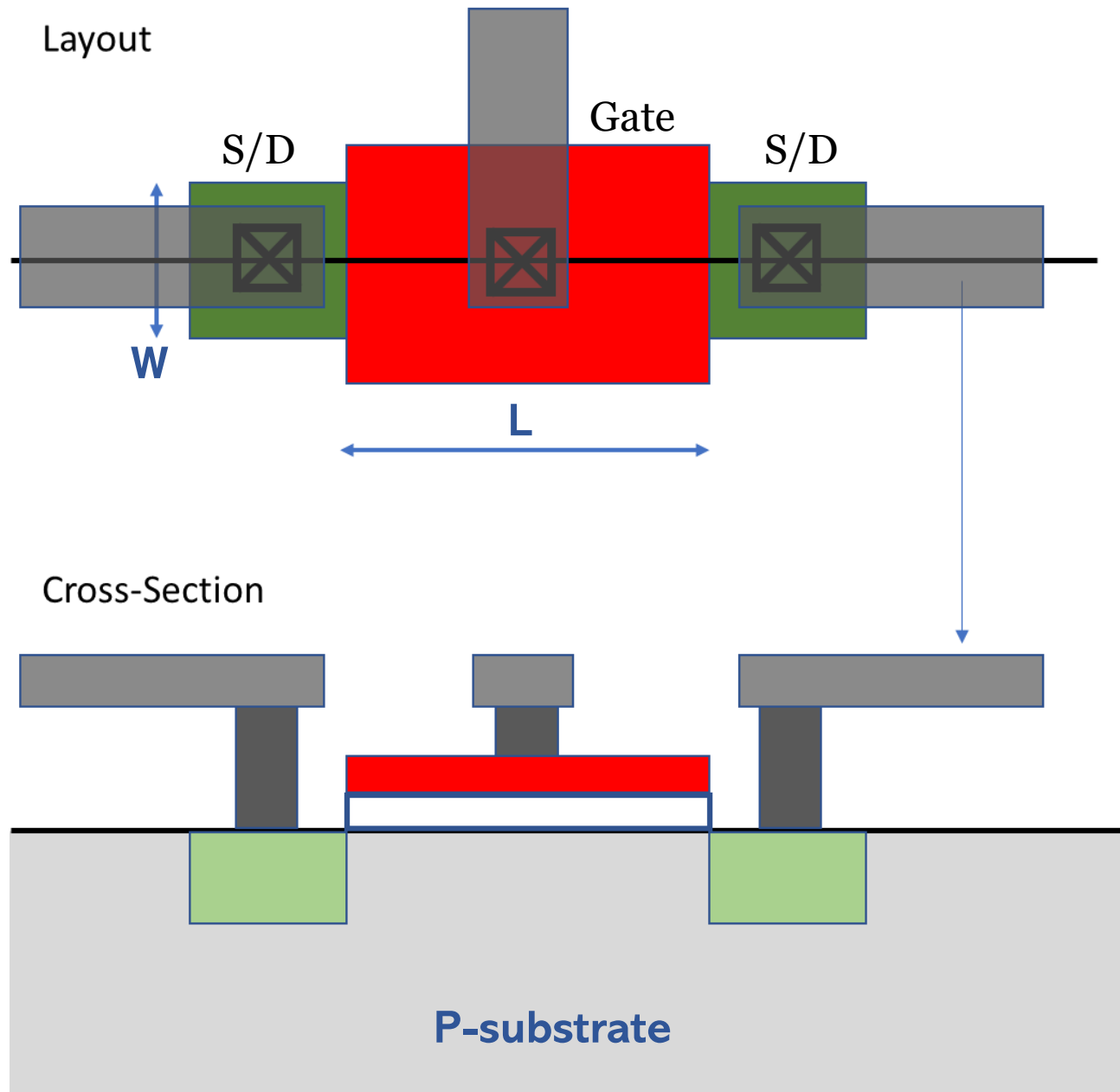
V_{GS}

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

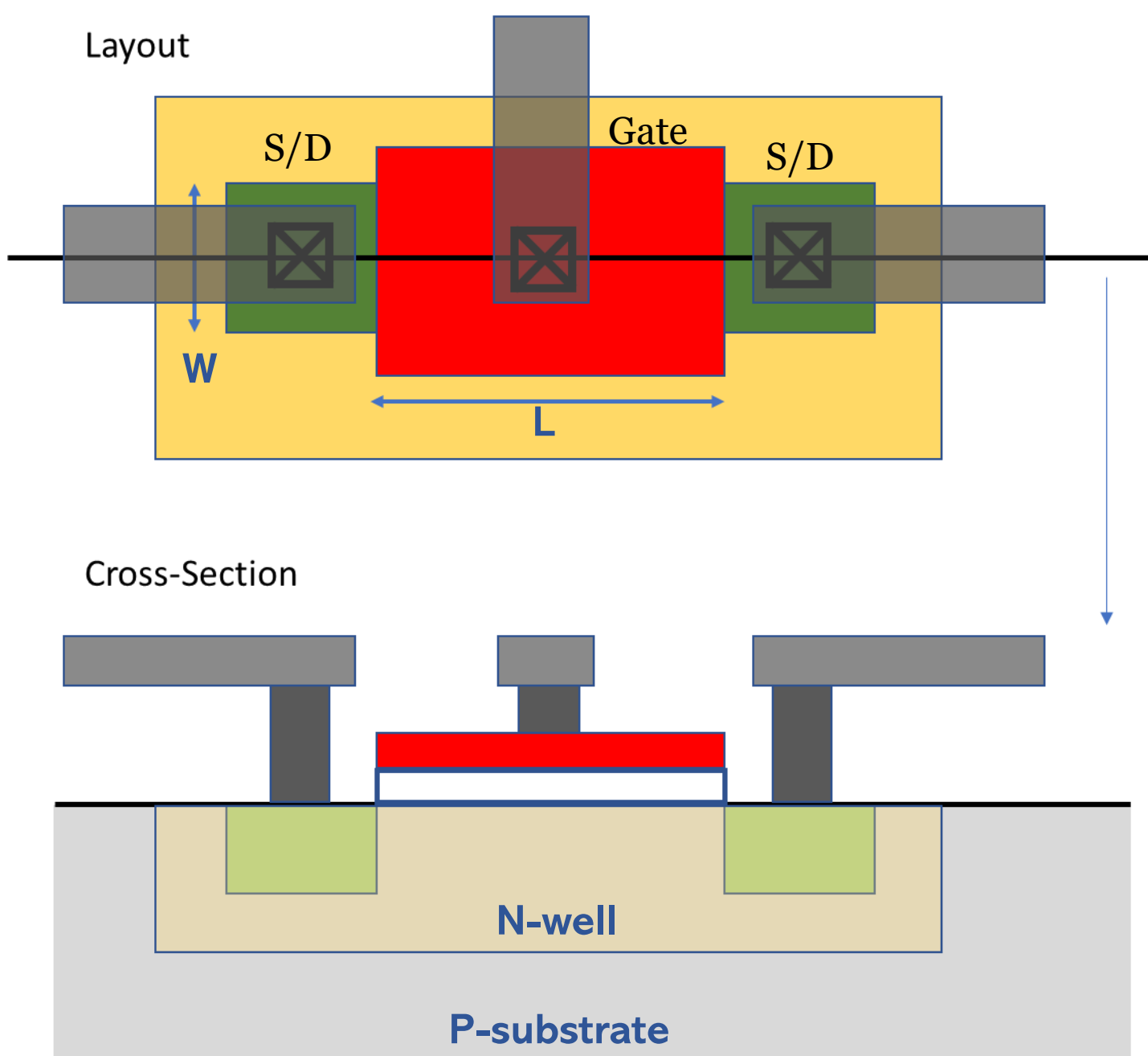
$$\sqrt{\frac{I_{DS}}{\frac{\mu_n C_{ox}}{2} \frac{W}{L}}} = V_{GS} - V_T$$

$$V_{GS} = V_T + \sqrt{\frac{I_{DS}}{C}}$$

► nMOS Transistor



► pMOS Transistor



I-V Equation for nMOS

$$I_D = 0$$

For $V_{GS} < V_T$

$$I_D(lin) = \frac{\mu_n C_{OX}}{2} \frac{W}{L} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

For $V_{GS} \geq V_T$
and $V_{DS} < V_{GS} - V_T$

$$I_D(lin) = \frac{\mu_n C_{OX}}{2} \frac{W}{L} [2(V_{GS} - V_T)V_{DS}]$$

For $V_{GS} \geq V_T$
and V_{DS} is small

$$I_D(sat) = \frac{\mu_n C_{OX}}{2} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

For $V_{GS} \geq V_T$
and $V_{DS} \geq V_{GS} - V_T$

I-V Equation for pMOS

$$I_D = 0$$

For $V_{GS} > V_T$

$$I_D(lin) = \frac{\mu_p C_{OX}}{2} \frac{W}{L} [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

For $V_{GS} \leq V_T$
and $V_{DS} > V_{GS} - V_T$

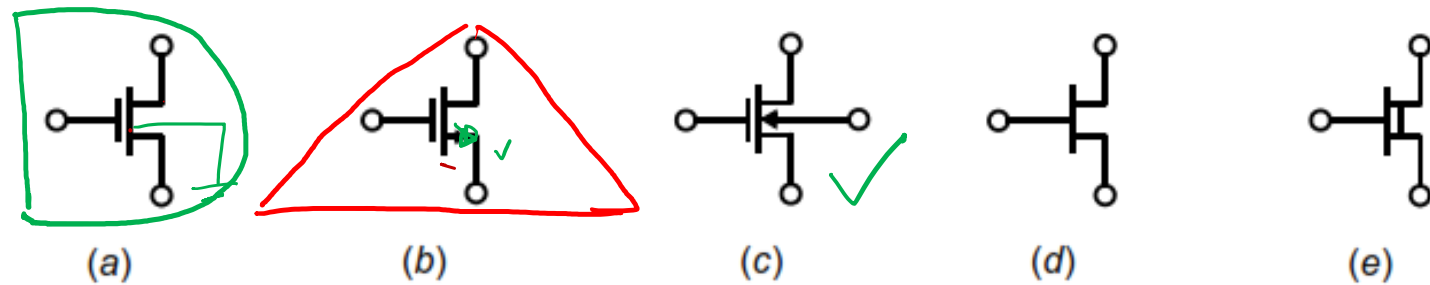
$$I_D(lin) = \frac{\mu_p C_{OX}}{2} \frac{W}{L} [2(V_{GS} - V_T)V_{DS}]$$

For $V_{GS} \leq V_T$
and V_{DS} is small

$$I_D(sat) = \frac{\mu_p C_{OX}}{2} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

For $V_{GS} \leq V_T$
and $V_{DS} \leq V_{GS} - V_T$

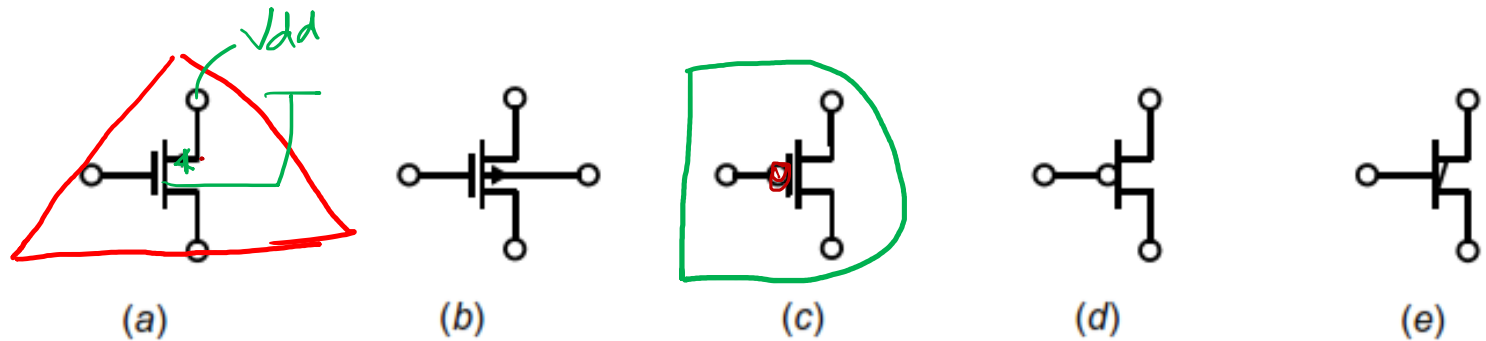
Structure and Operation of MOSFET



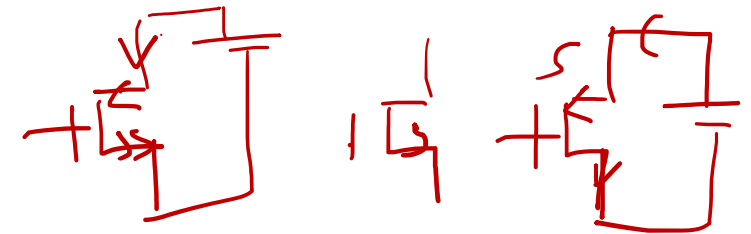
Commonly used symbols for n-channel transistors.

D → Symbol used in digital cks

△ → mostly used in analog cks



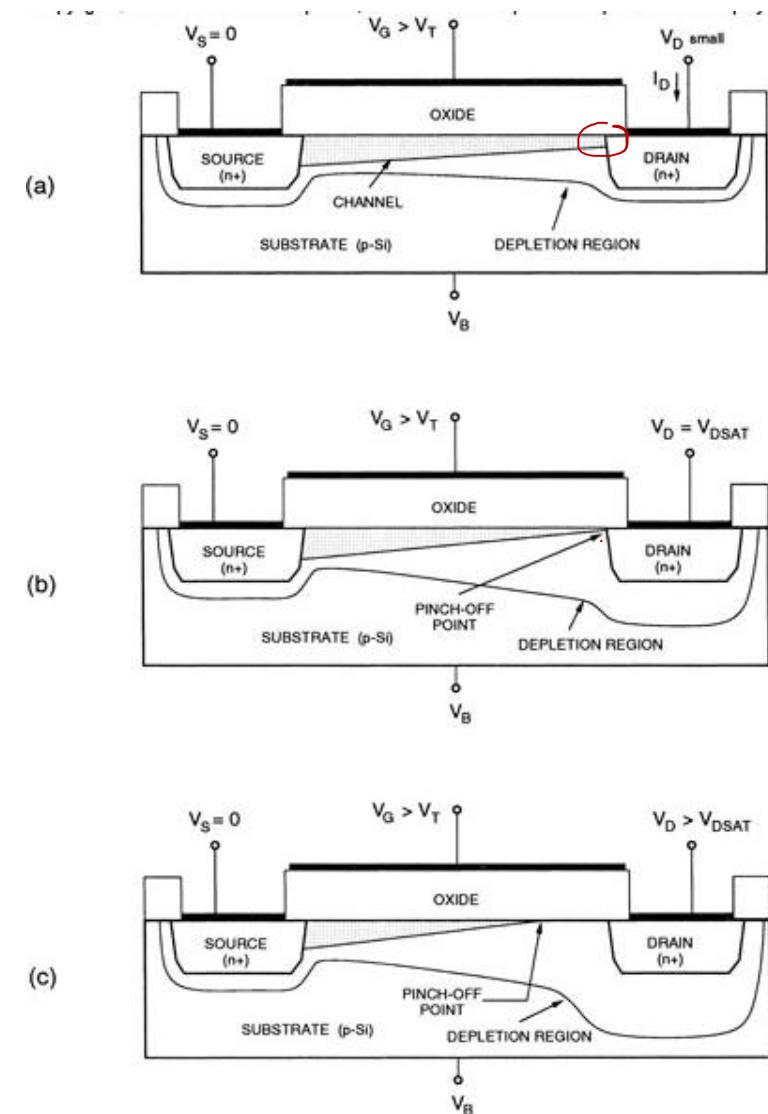
Commonly used symbols for p-channel transistors.



□ Circuit Symbols of MOSFETs

- The source is the n+(p+) region which has a lower (higher) potential than the other n+(p+) region in an n-channel (p-channel) MOSFET device.
- All the terminal voltage of the device are defined with respect to the source potential.

MOSFET Operation : A Quick View



➤ For $V_{GS} > V_{TO}$, $V_{DS} = 0$

- Drain current I_D equal to zero.

➤ For $V_{GS} > V_{TO}$, $0 < V_{DS} < V_{DSAT}$

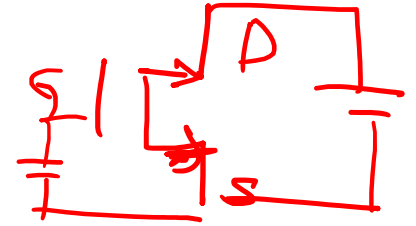
- Drain current I_D proportional to V_{DS}
- Called the linear mode (or linear region).

➤ For $V_{GS} > V_{TO}$, $V_{DS} = V_{DSAT}$

- Inversion charge at the drain is reduced to zero : pinch-off point.

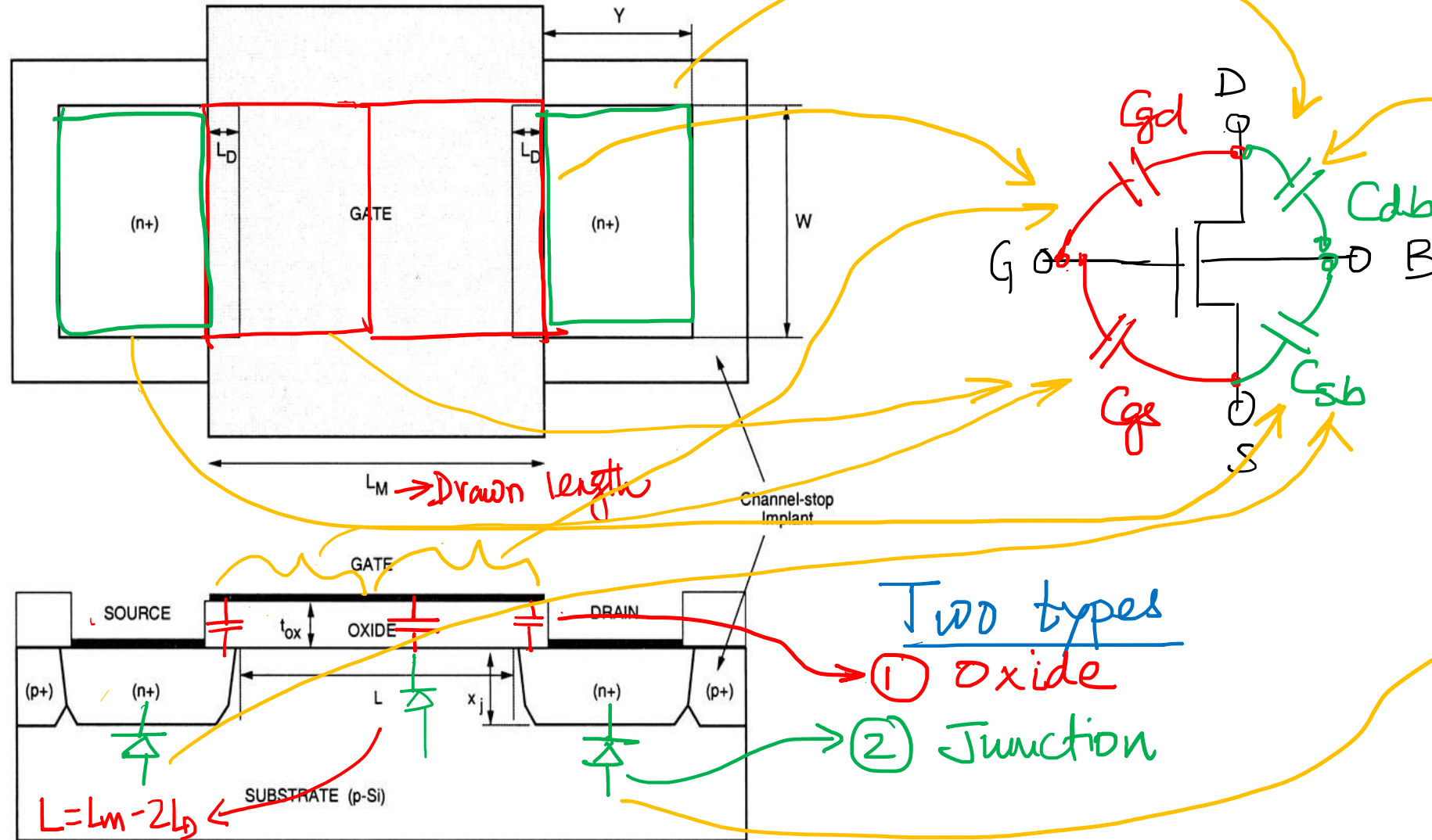
➤ For $V_{GS} > V_{TO}$, $V_{DS} > V_{DSAT}$

- A depleted surface region forms adjacent to the drain and grows toward source.
- Called Saturation mode (or saturation region)

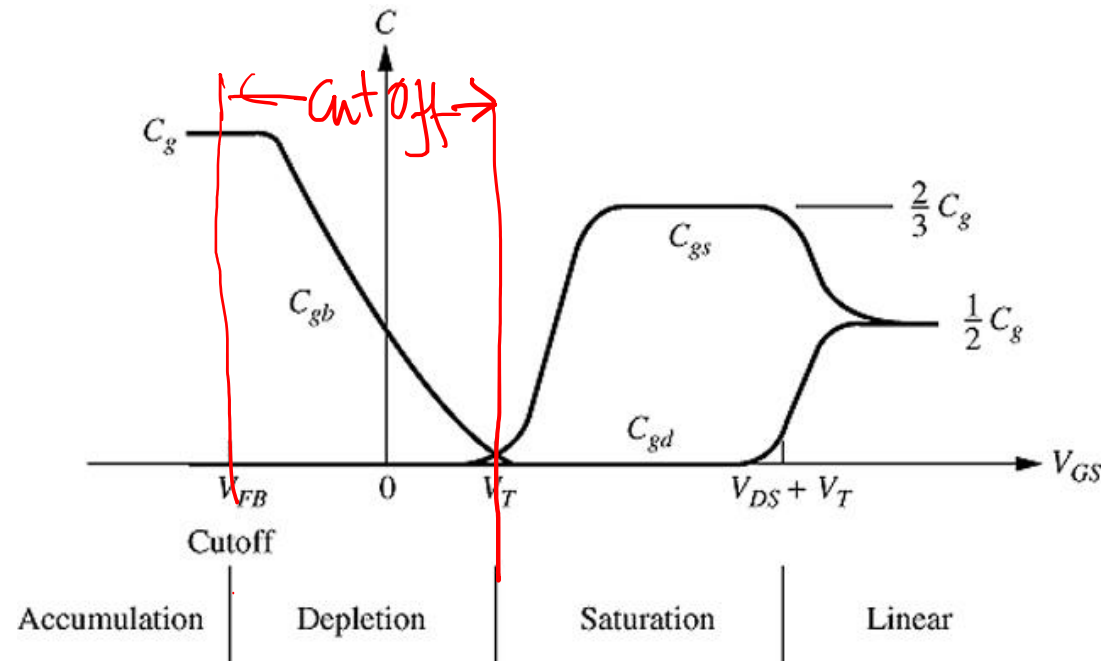
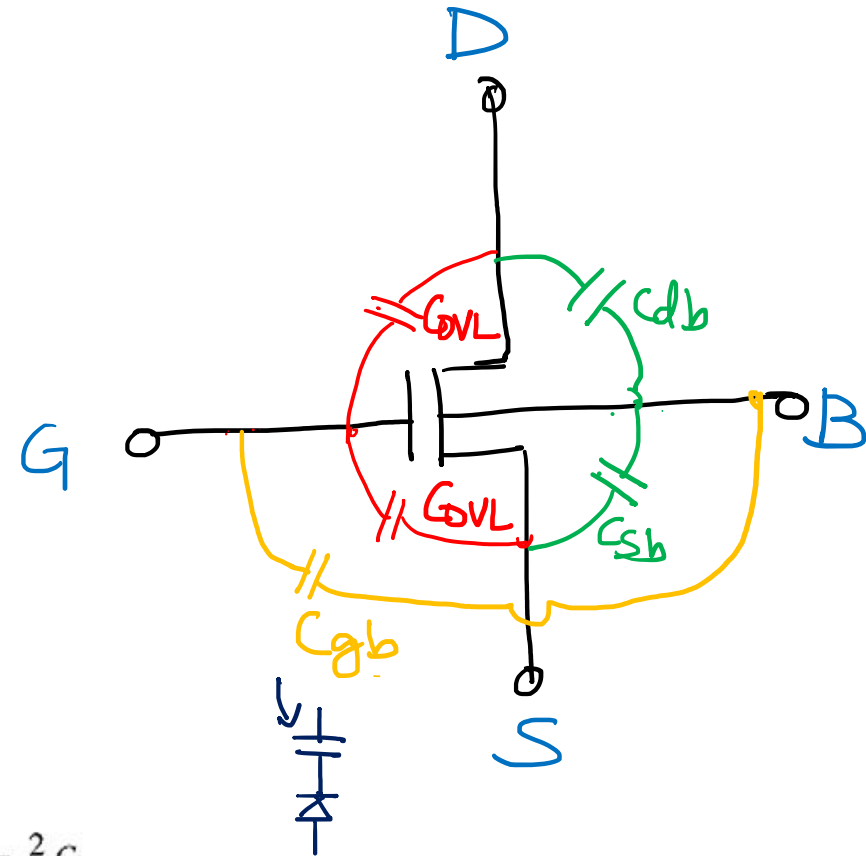
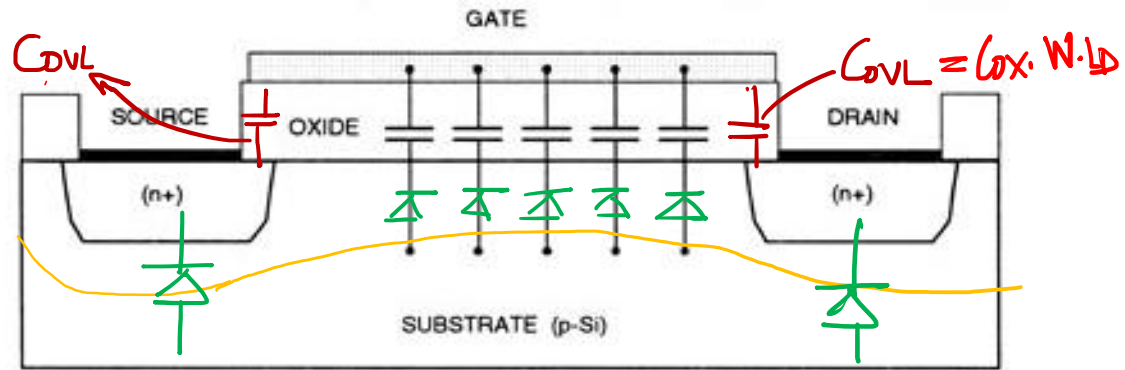


MOS Capacitance

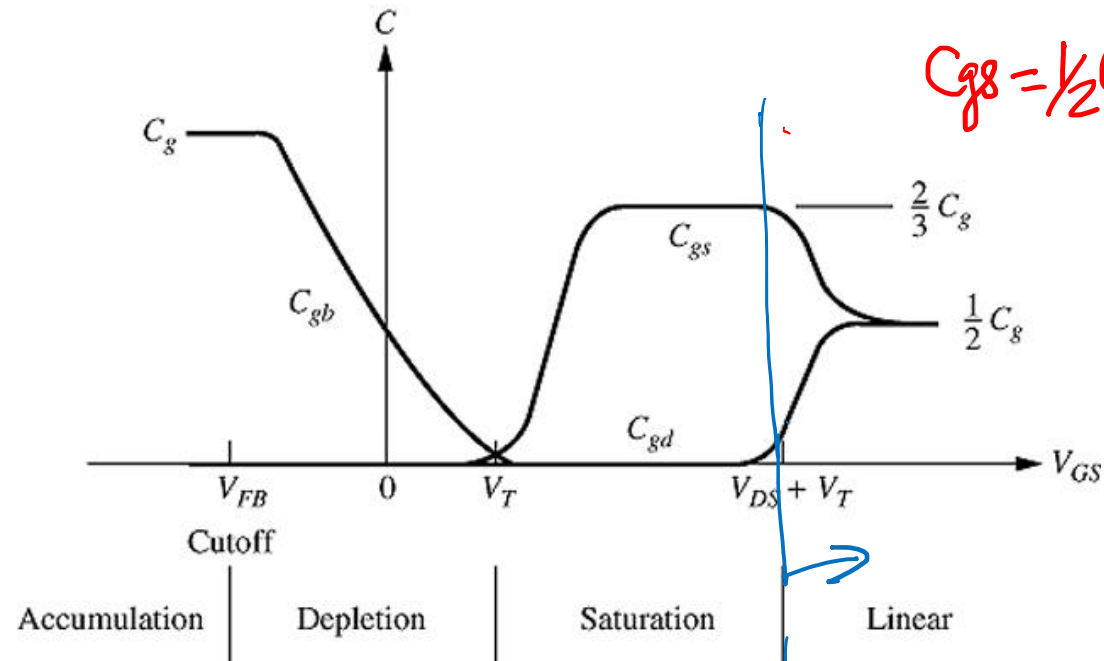
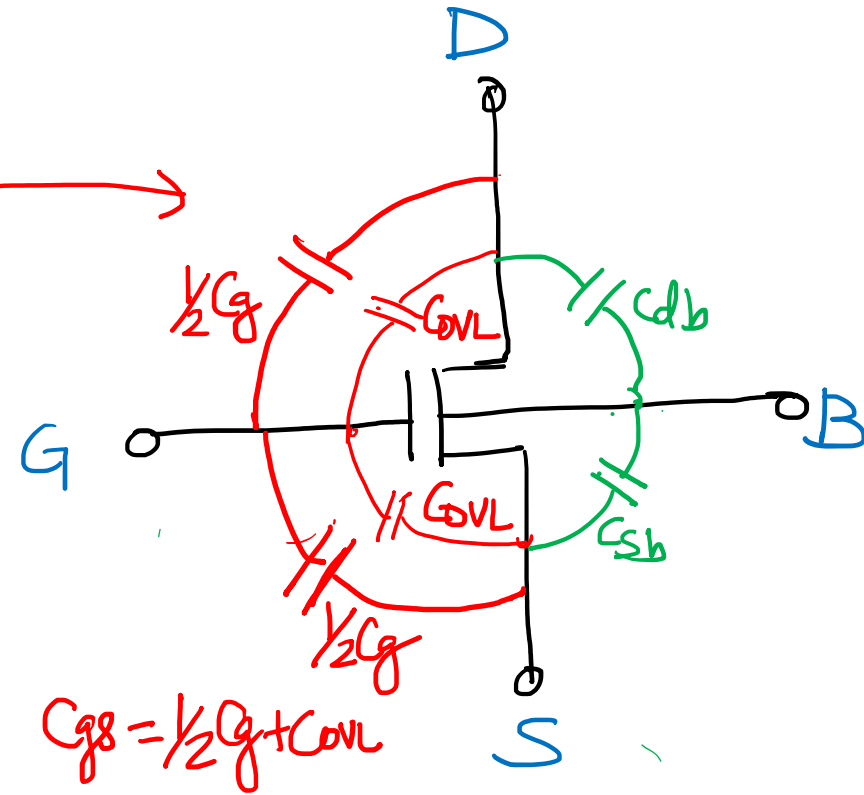
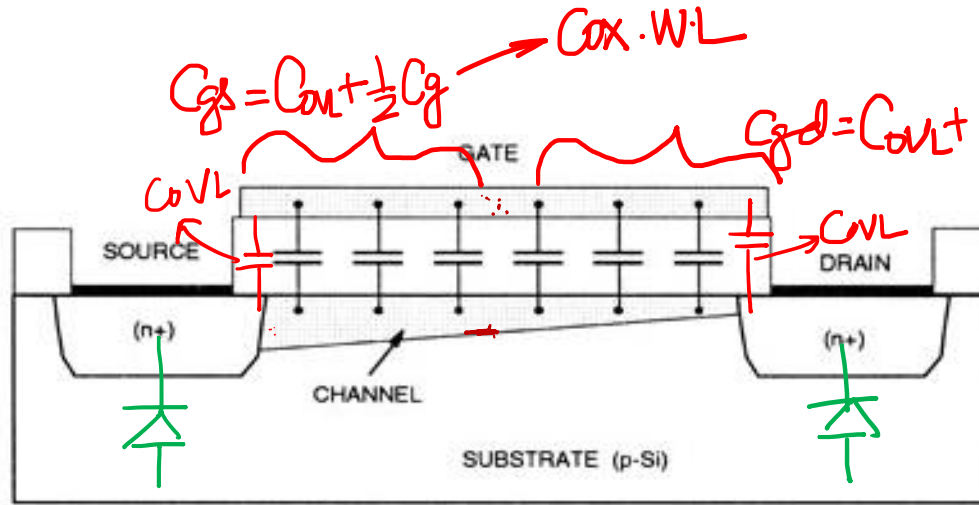
► MOS Capacitance



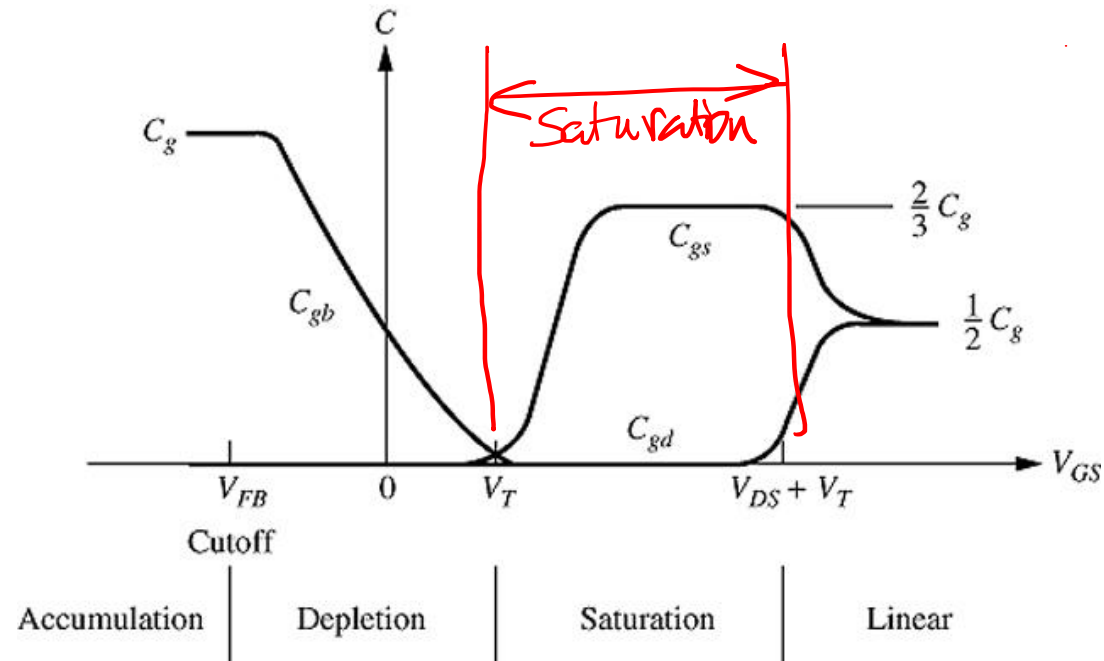
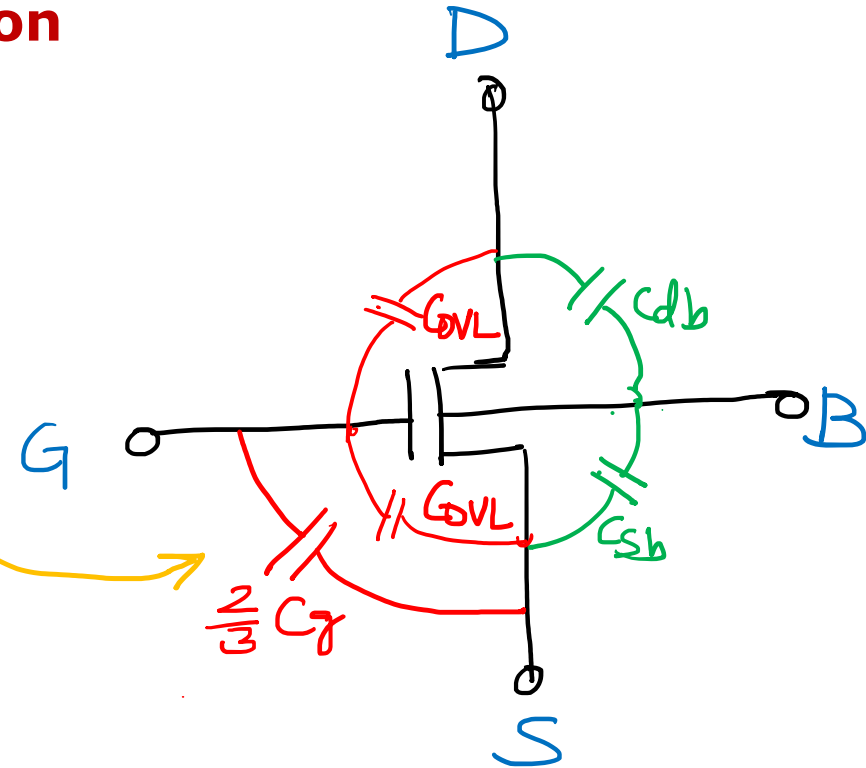
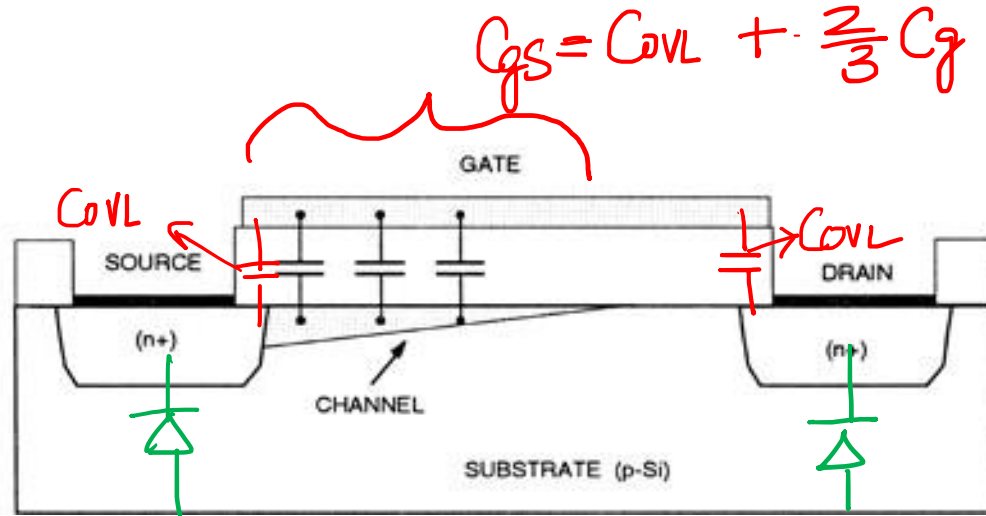
► Bias Dependent MOS Capacitance: Cut-Off

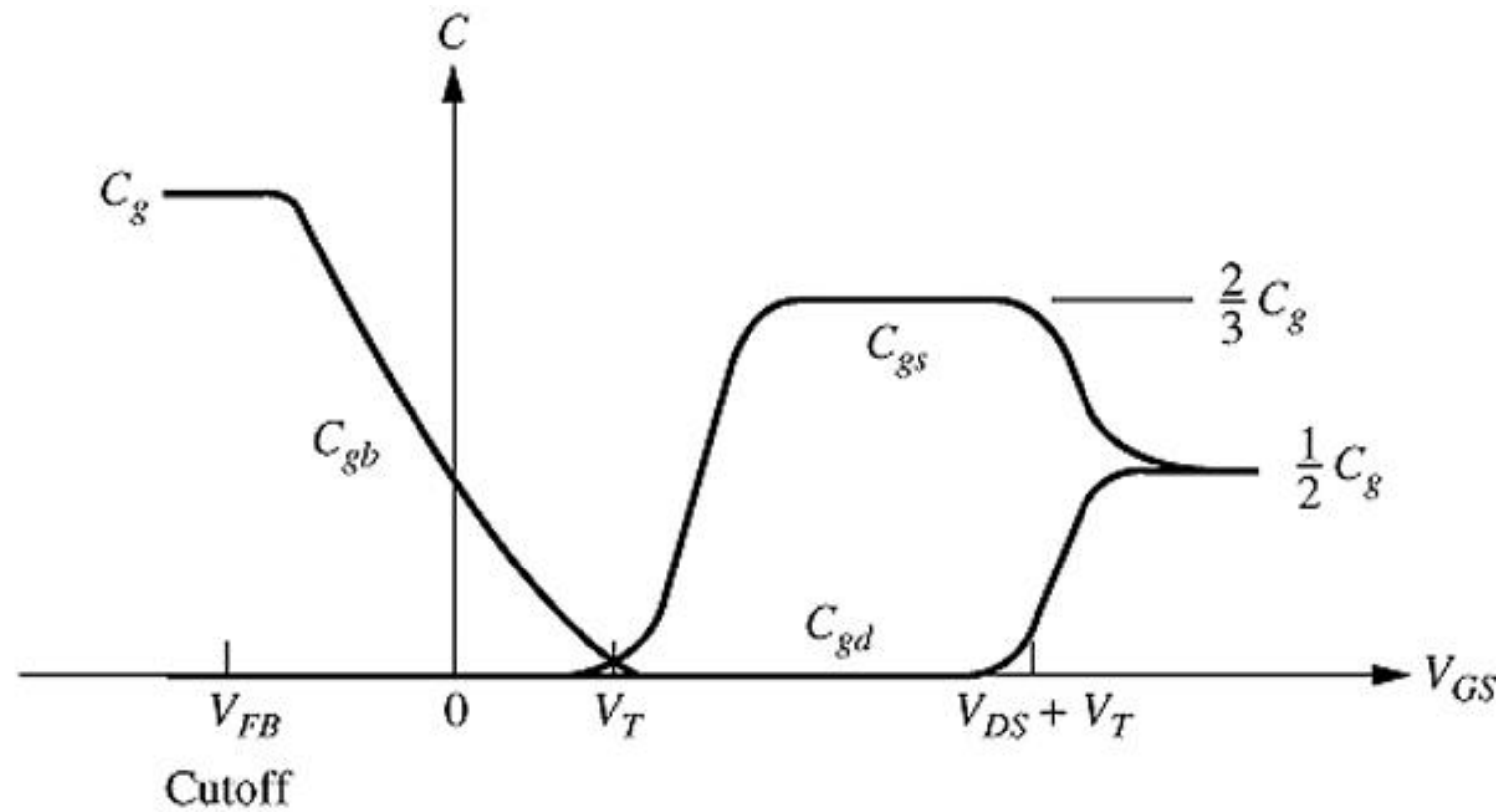


► **Bias Dependent MOS Capacitance: Linear**



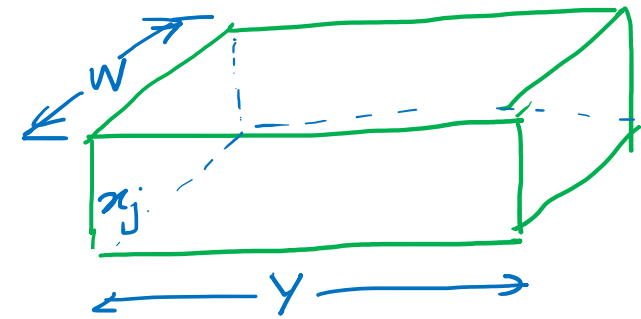
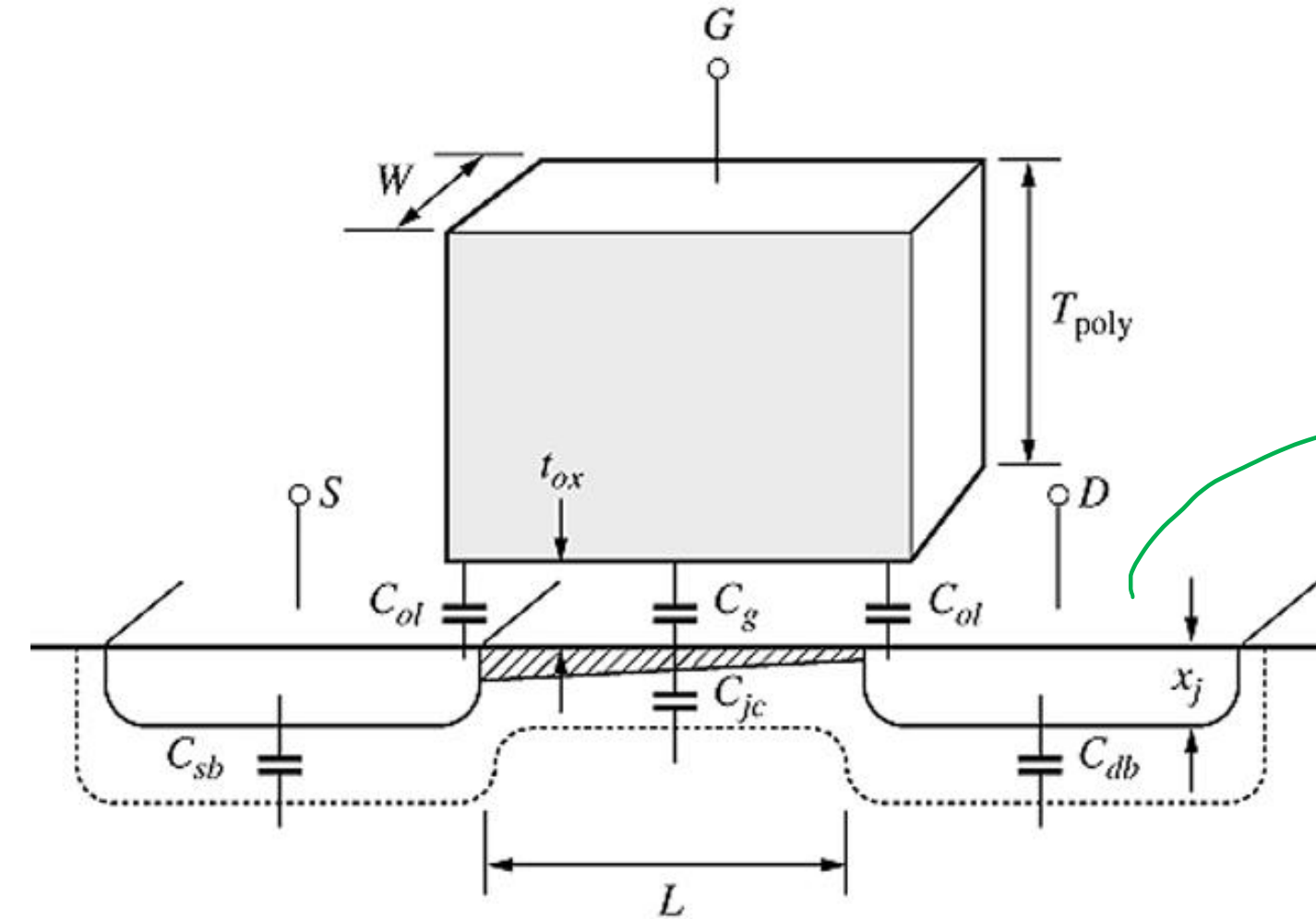
► Bias Dependent MOS Capacitance: Saturation





	Accumulation	Depletion	Saturation	Linear
C_{gs}	C_{ovL}	C_{ovL}	$C_{ovL} + \frac{2}{3} C_{ox}$	$C_{ovL} + \frac{1}{2} C_{ox}$
C_{gd}	C_{ovL}	C_{ovL}	C_{ovL}	$C_{ovL} + \frac{1}{2} C_{ox}$
C_{sb}, C_{sb}	C_j	C_j	C_j	C_j
C_{gb}	C_{ox}	$C_{ox} \parallel C_{dep}$	0	0

MOS Capacitance



$$C_g = C_{j0} W Y + C_{jsw} (2Y + W)$$